

Chapter 27

Introducing Young tableaux by examples

Given the highest weight of an irreducible representation (irrep) of a simple Lie algebra, one can construct the whole multiplet of states by acting with the lowering operators associated with the simple roots, and also the Clebsch-Gordan (CG) decomposition of the tensor product of two irreps can be constructed by acting on the product of the two highest weight states. However, the procedure is laborious, and tensor methods yield the results in a much simpler and faster way. **Tensor methods can be visualized by Young tableaux.** In this section we give an introduction to Young tableaux by working out a series of simple cases. We begin with irreps of $SU(N)$ where we use the simple formulas for the dimensions of irreps and for the CG decomposition. Then we shall discuss Young tableaux for the orthogonal groups. New here are the Young tableaux for spinor irreps. Using the isomorphisms between some low $SU(N)$ and $SO(N)$ groups, we can check some of the results of Young tableaux. Finally we shall discuss Young tableaux for the symplectic group $USp(2N)$, and also here some isomorphisms allow us to make checks. Neither $SU(N)$ nor $USp(2N)$ have spinor irreps.

$SU(2)$. We begin with the simplest case, $SU(2)$. The vector (defining, fundamental) irrep is two-dimensional and it is represented by a box

$$\boxed{}_{\mathbf{2}} : \text{the spin } j = \frac{1}{2} \text{ irrep} \quad (27.1)$$

We write below Young tableaux their dimensions. From quantum mechanics we know how to add angular momenta. If $j_1 \geq j_2$ one has $j_1 \otimes j_2 = j_1 + j_2, j_1 + j_2 - 1, \dots, j_1 - j_2$. A spin j irrep is obtained by taking the product of $2j$ spin $\frac{1}{2}$ irreps and symmetrizing in all indices of the tensor

$$\boxed{}\boxed{}\boxed{}\boxed{}\boxed{} : \text{the spin } j \text{ irrep from } 2j \text{ boxes: } t^{(\alpha_1 \dots \alpha_{2j})} \quad (27.2)$$

The symbol $(\dots)$ indicates total symmetrization of the indices inside the parenthesis, and the symbol $[\dots]$ denotes total antisymmetrization. For $SO(N)$ groups we shall use the symbol $((\dots))$ to denote total symmetrization and tracelessness, and for $USp(2N)$ groups the symbol $[[\dots]]$

denotes total antisymmetrization and symplectic-tracelessness. For example in $SO(N)$ $t^{((ij))} = t^{(ij)} - \frac{1}{N}\delta^{ij}t$ with $t = \delta_{kl}t^{kl}$, and in $USp(2N)$ $t^{[[ij]]} = t^{[ij]} - \frac{1}{2N}\Omega^{ij}(\Omega_{kl}t^{kl})$ where $\Omega_{ij}\Omega^{ij} = 2N$.

Taking the product of two spin $\frac{1}{2}$ irreps yields a sum of a spin 1 and a spin 0 irrep.

$$\begin{array}{rclcl}
\psi^\alpha \chi^\beta & = & \psi^{(\alpha} \chi^{\beta)} & + & \psi^{[\alpha} \chi^{\beta]} \\
\text{spins: } \frac{1}{2} \otimes \frac{1}{2} & = & 1 & \otimes & 0 \\
\text{dims: } \mathbf{2} \times \mathbf{2} & = & \mathbf{3} & + & \mathbf{1} \\
\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \otimes \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} & = & \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array}
\end{array} \tag{27.3}$$

Since $\psi^{[\alpha} \chi^{\beta]} = \epsilon^{\alpha\beta}(\frac{1}{2}\epsilon_{\gamma\delta}\psi^\gamma\chi^\delta)$ and since $\epsilon_{\gamma\delta}\psi^\gamma\chi^\delta$ is a singlet (a scalar, it does not transform) because $\epsilon_{\gamma\delta}$ is an invariant tensor, we replace $\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array}$ by a dot $\bullet$.

$$\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} = \bullet \tag{27.4}$$

For the same reason we replace in $SU(N)$ a column with N boxes by a dot (we omit the dot if it is part of a Young tableaux).

$$\left. \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} \right\} N = \bullet; \quad \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} = \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \tag{27.5}$$

We can next add two spin 1 irreps, giving a sum of spin 2, 1, 0 irreps

$$\begin{array}{ccccccc}
\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} & \otimes & \begin{array}{|c|c|} \hline a & a \\ \hline \end{array} & = & \begin{array}{|c|c|c|c|} \hline \square & \square & a & a \\ \hline \end{array} & \oplus & \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline a & & \\ \hline \end{array} & \oplus & \begin{array}{|c|c|} \hline \square & \square \\ \hline a & a \\ \hline \end{array} \\
\mathbf{3} & \times & \mathbf{3} & = & \mathbf{5} & + & \mathbf{3} & + & \mathbf{1}
\end{array} \tag{27.6}$$

Two boxes in one column give a singlet of $SU(2)$ and can be omitted. So for example

$$\begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} = \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \quad \text{in } SU(2). \tag{27.7}$$

All these results for $SU(2)$ are familiar from quantum mechanics.

$SU(3)$. The next simplest group is $SU(3)$. Now the vector irrep is 3-dimensional, indicated by writing $\mathbf{3}$ and depicted by one box

$$\begin{array}{|c|} \hline \square \\ \hline \end{array} : v^\alpha \text{ of } SU(3). \tag{27.8}$$

Next we consider the complex conjugate of the vector v^α , which we denote by a vector w_α with a covariant index: $(v^\alpha)^* = w_\alpha$, and which corresponds to two boxes

$$\begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \quad t^{\alpha\beta} = \epsilon^{\alpha\beta\gamma} w_\gamma, \quad w_\gamma = \frac{1}{2}\epsilon_{\gamma\alpha\beta} t^{\alpha\beta} \tag{27.9}$$

We used that $\epsilon^{\alpha\beta\gamma}w_\beta = v^{\alpha\beta}$ is a two-index antisymmetric tensor $t^{[\alpha\beta]}$ in $SU(3)$. Let's do some exercises

$$\begin{array}{rclcl}
\begin{array}{|c|} \hline \square \\ \hline \end{array} & \otimes & \begin{array}{|c|} \hline \square \\ \hline \end{array} & = & \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \\
\mathbf{3}^* & \times & \mathbf{3} & = & \mathbf{8} + \mathbf{1} \\
w_\alpha & \times & v^\beta & = & w_\alpha v^\beta - \frac{1}{3}\delta_\alpha^\beta(v^\gamma w_\gamma) + \frac{1}{3}\delta_\alpha^\beta(v^\gamma w_\gamma)
\end{array} \tag{27.10}$$

Recall $\mathbf{3}^* = w_\alpha = \frac{1}{2}\epsilon_{\alpha\beta\gamma}v^{\beta\gamma}$. Next we consider the product of two tensors v^α and w^β

$$\begin{array}{rclcl}
\begin{array}{|c|} \hline \square \\ \hline \end{array} & \otimes & \begin{array}{|c|} \hline \square \\ \hline \end{array} & = & \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \\
\mathbf{3} & \times & \mathbf{3} & = & \mathbf{6} + \mathbf{3}^* \\
u^\alpha & \times & v^\beta & = & \frac{1}{2}(u^\alpha v^\beta + u^\beta v^\alpha) + \frac{1}{2}(u^\alpha v^\beta - u^\beta v^\alpha) \\
& & & = & u^{(\alpha}v^{\beta)} + u^{[\alpha}v^{\beta]}
\end{array} \tag{27.11}$$

A symmetric tensor $t^{\alpha\beta} = t^{\beta\alpha}$ of $SU(3)$ has 6 components. We go on

$$\begin{array}{rclcl}
\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} & \otimes & \begin{array}{|c|} \hline \square \\ \hline \end{array} & = & \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \\
\mathbf{6} & \times & \mathbf{3} & = & \mathbf{10} + \mathbf{8}
\end{array} \tag{27.12}$$

A symmetric tensor with k indices in $SU(N)$ (so the indices can take on N values) has $\binom{N+k-1}{k}$ components; this explains the **10**. The **8** is the adjoint rep of $SU(3)$, its highest weight is $\vec{\mu}_{HW}(\mathbf{8}) = \vec{\mu}_{FW}^{(1)} + \vec{\mu}_{FW}^{(N-1)}$, and since $\vec{\mu}_{FW}^{(k)}$ corresponds to a k -index antisymmetric tensor, the tableaux for the **8** becomes clear. We leave as an exercise

$$\begin{array}{rclcl}
\begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} & \otimes & \begin{array}{|c|} \hline \square \\ \hline \end{array} & = & \begin{array}{|c|c|c|c|} \hline \square & \square & \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} \\
\mathbf{10} & \times & \mathbf{3} & = & \mathbf{15} + ?
\end{array} \quad \text{Clearly } \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array} = \mathbf{15} \tag{27.13}$$

By matching dimensions we discover that the irreducible tensor $t_\gamma^{\alpha\beta} = \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \end{array}$ has 15 components. That makes sense: $t^{\alpha\beta} = t^{\beta\alpha}$ has 6 components, so $t^{\alpha\beta}v_\gamma$ has 18 components, but subtracting the trace $t^{\alpha\beta}v_\beta$ removes 3 components, leaving 15 components.

Next consider the product of two irreps which each have more than one box. For example

$$\begin{array}{rclcl}
\begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} & \otimes & \begin{array}{|c|} \hline a \\ \hline b \\ \hline \end{array} & = & \left(\begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \end{array} \right) \otimes \begin{array}{|c|} \hline b \\ \hline \end{array} = \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & a \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & b \\ \hline \end{array} \oplus \begin{array}{|c|c|c|} \hline \square & \square & b \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline a \\ \hline b \\ \hline \end{array} \\
\mathbf{6} & \times & \mathbf{3}^* & = & \dots = \mathbf{15} + \mathbf{3}
\end{array} \tag{27.14}$$

The rule is: add first box a in all possible ways, then box b in all possible ways, but reading from right to left and top to bottom, the number of boxes with a should always be larger or equal to the number boxes with b .

Using that $\mathbf{6}^* = (\square\square)^* = \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}$ we find

$$\begin{array}{ccccc} \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} & \otimes & \square & = & \begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix} \oplus \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \\ \mathbf{6}^* & \times & \mathbf{3} & = & \mathbf{15}^* + \mathbf{3}^* \end{array} \quad \text{where } \left(\begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix} \right)^* = \begin{smallmatrix} \square & \square & \square \\ \square & \square & \square \end{smallmatrix} \quad (27.15)$$

We used that $\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} = \begin{smallmatrix} \square \\ \square \end{smallmatrix} = \mathbf{3}^*$. Finally

$$\begin{array}{ccccc} \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} & \otimes & \begin{smallmatrix} a & a \end{smallmatrix} & = & \begin{smallmatrix} \square & \square & a & a \\ \square & \square & a & a \end{smallmatrix} \oplus \begin{smallmatrix} \square & \square & a \\ \square & \square & a \end{smallmatrix} \oplus \begin{smallmatrix} \square & \square & a \\ a & \square & \square \end{smallmatrix} \oplus \begin{smallmatrix} \square & \square \\ a & a \end{smallmatrix} \oplus \begin{smallmatrix} \square & \square \\ a & a \end{smallmatrix} \oplus \begin{smallmatrix} \square & \square \\ a & a \end{smallmatrix} \\ \mathbf{6}^* & \times & \mathbf{6} & = & ? + \mathbf{8} + \mathbf{1} \end{array} \quad (27.16)$$

Another rule is: no two a boxes in the same column. By matching dimensions we find out

$$\begin{smallmatrix} \square & \square & \square & \square \\ \square & \square & \square & \square \end{smallmatrix} = \mathbf{27}. \quad (27.17)$$

In tensor notation this corresponds to

$$u_{(\alpha\beta)}v^{(\gamma\delta)} = t_{\alpha\beta}^{\gamma\delta} + p\delta_{(\alpha}^{\gamma}(u \cdot v)_{\beta)}^{\delta)} + q\delta_{(\gamma}^{(\alpha}\delta_{\delta)}^{\beta)}(u \cdot v) \quad (27.18)$$

where p and q are coefficients (numbers) and $t_{\alpha\beta}^{\gamma\delta}$ is symmetric in $\alpha\beta$ and $\gamma\delta$ and traceless in $\alpha\beta$, $\alpha\delta$, $\beta\gamma$ and $\beta\delta$, and $(u \cdot v)_{\beta}^{\delta} = u_{\alpha\beta}v^{\gamma\delta}\delta_{\alpha}^{\gamma}$ while $u : v = u_{\alpha\beta}v^{\alpha\beta}$. By taking traces on both sides one may fix the coefficients p and q .

$SU(4)$. We move on to $SU(4)$. We have the following irreps of low dimensions

$$\begin{array}{ccccccccccc} \square, & \square\square, & \begin{smallmatrix} \square \\ \square \end{smallmatrix}, & \begin{smallmatrix} \square \\ \square \\ \square \end{smallmatrix}, & \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}, & \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}, & \square\square\square, & \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}, & \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix}, & \dots \\ \mathbf{4} & \mathbf{10} & \mathbf{6} & \mathbf{4}^* & \mathbf{10}^* & \mathbf{15} & \mathbf{20} & \mathbf{20}' & \mathbf{20}'' \end{array} \quad (27.19)$$

The **15** is the adjoint rep, corresponding to a traceless tensor t_{β}^{α} , as follows

$$\begin{array}{ccccc} \begin{smallmatrix} \square \\ \square \\ \square \end{smallmatrix} & \otimes & \square & = & \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \oplus \begin{smallmatrix} \square \\ \square \\ \square \end{smallmatrix} \\ \mathbf{4}^* & \times & \mathbf{4} & = & \mathbf{15} + \mathbf{1} \end{array} \quad \begin{array}{l} v^{\alpha}w_{\beta} = t_{\beta}^{\alpha} + \frac{1}{4}\delta_{\beta}^{\alpha}(v \cdot w) \\ t_{\beta}^{\alpha} = v^{\alpha}w_{\beta} - \frac{1}{4}\delta_{\beta}^{\alpha}(v \cdot w) \end{array} \quad (27.20)$$

Let us do $\mathbf{10} \times \mathbf{4}$ in two ways

$$\begin{array}{ccccc} \square\square & \otimes & \square & = & \square\square\square \oplus \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \\ \mathbf{10} & \times & \mathbf{4} & = & \mathbf{20} + ? \end{array} \quad \text{Clearly } \begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} = \mathbf{20}''^* = \left(\begin{smallmatrix} \square & \square \\ \square & \square \end{smallmatrix} \right)^* \quad (27.21)$$

Now the other way around

$$\begin{array}{ccccccc}
\begin{array}{|c|} \hline \square \\ \hline \end{array} & \otimes & \begin{array}{|c|c|} \hline a & a \\ \hline \end{array} & = & \begin{array}{|c|c|c|} \hline & a & a \\ \hline \end{array} & \oplus & \begin{array}{|c|c|} \hline & a \\ a & \\ \hline \end{array} \oplus \begin{array}{|c|} \hline a \\ \hline \end{array} \\
\mathbf{4} & \times & \mathbf{10} & = & \mathbf{20} & + & \mathbf{20}''^*
\end{array} \tag{27.22}$$

The last diagram is to be discarded since there should not be two a -boxes in the same column.

$SO(N)$. Next we study orthogonal groups. At the level of Young tableaux for tensors we can treat $SO(2N)$ and $SO(2N+1)$ the same way, but recall that $SO(2N)$ has two spinor irreps which we denote by $\times$ and $+$, whereas $SO(2N+1)$ has only one spinor irrep which we denote by $\times$. Also Young tableaux for $SO(2N)$ and $SO(2N+1)$ are at most N boxes deep because an antisymmetric tensor with more than N boxes can be written as $\epsilon^{\mu_1 \dots \mu_{2N}}$ times a tensor with less than N boxes. A column with $2N$ boxes yields again a scalar ($t^{\mu_1 \dots \mu_{2N}}$ is invariant since it is proportional to the invariant tensor $\epsilon^{\mu_1 \dots \mu_{2N}}$). We shall use the isomorphisms between $SO(3)$ and $SU(2)$, and between $SO(6)$ and $SU(4)$, to clarify the results for $SO(N)$.

$SO(3)$. Denoting the **3** by $\square$ and the spinor irrep **2** by $\times$ we get

$$\begin{array}{ccccccc}
\square & \otimes & \square & = & (\square\square \oplus \bullet) & \oplus & \begin{array}{|c|} \hline \square \\ \hline \end{array} \\
\mathbf{3} & \times & \mathbf{3} & = & (\mathbf{5} + \mathbf{1}) & + & \mathbf{3} \\
u^\alpha & \times & v^\beta & = & (u^{((\alpha} v^{\beta))}) & + & \frac{1}{3} \delta^{\alpha\beta} u \cdot v + \frac{1}{2} \epsilon^{\alpha\beta\gamma} (\vec{u} \times \vec{v})_\gamma
\end{array} \tag{27.23}$$

Since $(\vec{u} \times \vec{v})_\gamma = \epsilon_{\gamma\alpha\beta} u^\alpha v^\beta$, the last term is equal to $u^{[\alpha} v^{\beta]}$. Here, $\begin{array}{|c|} \hline \square \\ \hline \end{array} = \square$ and $\bullet$ is the trace (a singlet). Recall that the pair of parenthesis $((\dots))$ denotes a symmetric and traceless tensor.

Now the spinor irrep. Recall $SU(2) \simeq SO(3)$ so $\square$ of $SU(2)$ is $\times$ of $SO(3)$.

$$\begin{array}{lcl}
SO(3): & \begin{array}{ccc} \square & \otimes & \times \\ \mathbf{3} & \times & \mathbf{2} \\ v^\mu & & \psi^\alpha \end{array} & = & \begin{array}{ccc} \begin{array}{|c|} \hline \times \\ \hline \end{array} & \oplus & \times \\ \mathbf{4} & + & \mathbf{2} \end{array} \\
& & & & v^\mu \psi^\alpha - \frac{1}{3} (\gamma^\mu)^\alpha_\beta v \cdot \gamma \psi^\beta & + & \frac{1}{3} (\gamma^\mu)^\alpha_\beta v \cdot \gamma \psi^\beta \\
SU(2): & \begin{array}{ccc} \square\square & \otimes & \square \\ (j=1) & \otimes & (j=\frac{1}{2}) \end{array} & = & \begin{array}{ccc} \square\square\square & \oplus & \square \\ (j=\frac{3}{2}) & \oplus & (j=\frac{1}{2}) \end{array}
\end{array} \tag{27.24}$$

The vector-spinor $v^\mu \psi^\alpha$ with $\mu = 1, 2, 3$ the vector index and $\alpha = 1, 2$ the spinor index, is made irreducible by subtracting the “gamma-trace”, namely $t^{\mu\alpha} = v^\mu \psi^\alpha - \frac{1}{3} (\gamma^\mu)^\alpha_\beta v \cdot \gamma \psi^\beta$ where $v \cdot \gamma \psi^\beta = v^\mu (\gamma_\mu)^\beta_\gamma \psi^\gamma$. The γ^μ are the three Pauli matrices. One may check that $t^{\mu\alpha}$ is gamma-traceless: $(\gamma_\mu)^\alpha_\beta t^{\mu\beta} = 0$. Another example where we compare the Young tableaux of $SO(3)$ and $SU(2)$ is as

follows

$$\begin{aligned}
SO(3) : \quad & \times \quad \otimes \quad \times \quad = \quad \square \quad \oplus \quad \bullet \\
& \mathbf{2} \quad \times \quad \mathbf{2} \quad = \quad \mathbf{3} \quad + \quad \mathbf{1} \\
SU(2) : \quad & \square \quad \otimes \quad \square \quad = \quad \square\square \quad \oplus \quad \begin{array}{c} \square \\ \square \end{array} \\
& (j = \tfrac{1}{2}) \quad \otimes \quad (j = \tfrac{1}{2}) \quad = \quad (j = 1) \quad \oplus \quad (j = 0)
\end{aligned} \tag{27.25}$$

$SO(6)$. Having studied $SO(3)$, consider $SO(6)$. Since $SO(6) \simeq SU(4)$ we can try to match the irreps of $SO(6)$ and $SU(4)$. The first few are easy

$$\begin{aligned}
SU(4) : \quad & \square \quad \begin{array}{c} \square \\ \square \\ \square \end{array} \quad \begin{array}{c} \square \\ \square \end{array} \quad \square\square \quad \begin{array}{c} \square\square \\ \square \end{array} \quad \begin{array}{c} \square\square \\ \square\square \end{array} \\
& \mathbf{4} \quad \mathbf{4^*} \quad \mathbf{6} \quad \mathbf{10} \quad \mathbf{15} \quad \mathbf{20} \\
SO(6) : \quad & \times \quad + \quad \square \quad \begin{array}{c} \square \\ \square \\ \square \end{array} \quad \begin{array}{c} \square \\ \square \end{array} \quad \square\square
\end{aligned} \tag{27.26}$$

(For $\begin{array}{c} \square\square \\ \square \end{array}$ of $SU(4)$ we used the formula for the dimensions of a Young tableau of $SU(N)$: $\frac{4 \times 3 \times 5 \times 4}{2 \times 3 \times 1 \times 2} = 20$.) We note a regularity: an odd number of boxes of $SU(4)$ gives spinorial irreps of $SO(6)$, while an even number gives tensorial irreps. Let us check this with a more complicated spinor irrep.

$$\begin{aligned}
SO(6) : \quad & \square \quad \otimes \quad \times \quad = \quad \boxed{\times} \quad \oplus \quad + \\
& \mathbf{6} \quad \times \quad \mathbf{4} \quad = \quad \mathbf{?} \quad + \quad \mathbf{4^*} \\
SU(4) : \quad & \begin{array}{c} \square \\ \square \end{array} \quad \otimes \quad \square \quad = \quad \begin{array}{c} \square\square \\ \square \end{array} \quad \oplus \quad \begin{array}{c} \square \\ \square \\ \square \end{array} \\
& \mathbf{6} \quad \times \quad \mathbf{4} \quad = \quad \mathbf{20} \quad + \quad \mathbf{4^*}
\end{aligned} \tag{27.27}$$

Clearly $\boxed{\times} = \mathbf{20}$

($\dim \begin{array}{c} \square\square \\ \square \end{array} = \frac{4 \times 3 \times 5}{1 \times 3 \times 1} = 20$)

Note that we used the $+$ for $SO(6)$ to match the $\mathbf{4^*}$ in $SU(4)$. The spinorial irreps are clearly products of the lowest spinor irreps with tensor irreps. A vector-spinor $v^{\mu\alpha}$ can be decomposed into a “gamma-traceless” part $v^{\mu\alpha} - \frac{1}{6}\gamma^\mu(\gamma_\nu v^{\nu\alpha})$ and a “gamma-trace” part with $\gamma_\nu v^{\nu\alpha} = w^\alpha$. If v_μ^α is chiral, $\gamma_c v_\mu^\alpha = v_\mu^\alpha$, and $v_\mu^\alpha = w_\mu \psi^\alpha$ with ψ^α the $\times$ irrep, then $\gamma \cdot v^\alpha \equiv \gamma^\rho v_\rho^\alpha$ is antichiral, $\gamma_c \gamma \cdot v = -\gamma \cdot v$, (the $+$ irrep). Here γ_c is the chirality matrix $\gamma_c = \alpha \gamma^1 \cdots \gamma^6$ satisfying $\gamma_c^2 = I$.

Finally products of two spinor irreps

$$\begin{aligned}
& \times \quad \otimes \quad \times \quad = \quad \begin{array}{c} \square \\ \square \\ \square \end{array} \quad \oplus \quad \square \\
& \mathbf{4} \quad \times \quad \mathbf{4} \quad = \quad \mathbf{10} \quad + \quad \mathbf{6}
\end{aligned} \tag{27.28}$$

This requires and will get much more discussion.

$USp(2N)$. Finally the symplectic groups. The basic irrep (the defining rep) is the $2N$ denoted by $\square$. A few examples for $USp(4)$

$$\begin{array}{ccccc} \square, & \square\square, & \begin{array}{|c|} \hline \square \\ \hline \end{array} & & \\ \mathbf{4} & \mathbf{10} & \mathbf{5} & & \text{(take out the symplectic trace)} \\ t^\alpha & v^{(\alpha\beta)} & t^{[[\alpha\beta]]} & & \end{array} \quad (27.29)$$

Let's work out $\square \otimes \square$ for $USp(4)$

$$\begin{array}{ccccccc} \square & \otimes & \square & = & \square\square & \oplus & \left(\begin{array}{|c|} \hline \square \\ \hline \end{array} \oplus \bullet \right) \\ \mathbf{4} & \times & \mathbf{4} & = & \mathbf{10} & + & (\mathbf{5} + \mathbf{1}) \end{array} \quad (27.30)$$

The $\bullet$ comes from the symplectic trace. Another case

$$\begin{array}{ccccccc} \square\square & \otimes & \square & = & \square\square\square & \oplus & \left(\begin{array}{|c|} \hline \square\square \\ \hline \end{array} \oplus \square \right) \\ \mathbf{10} & \times & \mathbf{4} & = & \mathbf{20} & + & (\mathbf{16} + \mathbf{4}) \end{array} \quad (27.31)$$

The $\square$ is from the symplectic trace. Since $SO(5) \cong USp(4)$ we can do the same check as for $SO(6) \cong SU(4)$, namely identify irreps

$$\begin{array}{ccccccc} SO(5) : & \square & \times & \begin{array}{|c|} \hline \square \\ \hline \end{array} & \boxtimes & ? & \\ & \mathbf{5} & \mathbf{4} & \mathbf{10} & \mathbf{16} & \mathbf{20} & \\ USp(4) : & \begin{array}{|c|} \hline \square \\ \hline \end{array} & \square & \square\square & ? & \square\square\square & \end{array} \quad (27.32)$$

The $\mathbf{16}$ of $SO(5)$ is a vector-spinor (with in general $4 \times 5 = 20$ components) made gamma-traceless (which removes a spinor with 4 components, leaving $20 - 4 = 16$ components). Since we see again that an odd number of boxes in $USp(4)$ corresponds to spinorial irreps of $SO(5)$, the $\mathbf{16}$ of $USp(4)$ must have an odd number of boxes. The case $\square\square\square$ is ruled out since it gives a $\mathbf{20}$, and $\begin{array}{|c|} \hline \square \\ \hline \end{array}$ is not allowed since only at most $N = 2$ boxes are allowed in a column. Hence the only possibility is

$$\begin{array}{|c|} \hline \square\square \\ \hline \end{array} = \mathbf{16} \quad \text{of } USp(4). \quad (27.33)$$

and indeed we found this result before, see (27.31).

Finally we must identify the $\mathbf{20}$ of $SO(5)$. It should be a spinorial irrep because $\square\square\square$ of $USp(4)$ has an odd number of boxes. It cannot be $\boxtimes$ since that gave the $\mathbf{16}$. The next possibility is $\begin{array}{|c|} \hline \boxtimes \\ \hline \end{array}$, a tensor-spinor $t^{\mu\nu,\alpha} = -t^{\nu\mu,\alpha}$. A product of a tensor $t^{\mu\nu} = -t^{\nu\mu}$ with a spinor ψ^α gives a reducible $\mathbf{40}$, but the gamma-trace is

$$\gamma_\mu t^{\mu\nu,\alpha} = \psi^{\nu,\alpha} = \mathbf{20} = \mathbf{16} + \mathbf{4} \quad (27.34)$$

(The reducible vector-spinor $\psi^{\nu,\alpha}$ can in turn be decomposed into the two irreps $\mathbf{16} + \mathbf{4}$.) So the

dimension of the tensor-spinor $t^{\mu\nu,\alpha}$ of $SO(5)$ is $40 - (16 + 4) = 20$.

$$\begin{array}{|c|} \hline \times \\ \hline \times \\ \hline \end{array} = \mathbf{20} \quad \text{of } SO(5). \quad (27.35)$$

Comment. The formulae for the dimensions of all irreps of $SU(N)$, $SO(2N)$, $SO(2N + 1)$ and $USp(2N)$ are given in chapter 20.

27.1 CG decompositions of products of spinor irreps

We now study and derive the CG series for direct products of spinorial irreps. For the $SO(2N + 1)$ groups there is only one kind of fundamental spinor irreps which we denote by $\times$, but for the $SO(2N)$ groups there are two kinds, the chiral s -rep which we denote by $\times$, and the antichiral c -rep which we denote by $+$. We work through an example of each case.

$$\begin{aligned} SO(5) : & \quad \times \otimes \times \\ SO(6) : & \quad \times \otimes \times \text{ and } \times \otimes +. \end{aligned} \quad (27.36)$$

First we shall show that the product of two spinors ψ^α and χ^β of $SO(2N + 1)$ can be expanded in terms of the complete set of Dirac matrices as

$$\psi^\alpha \chi^\beta = \sum_k a_k (\gamma^{\mu_1 \dots \mu_k} C^{-1})^{\alpha\beta} (\psi^T C \gamma_{\mu_1 \dots \mu_k} \chi) \quad (27.37)$$

where a_k are constants, which we shall determine. Then we consider the case of $SO(6)$.

In $SO(5)$ spinors have 4-components. The product $\psi^\alpha \chi^\beta$ of two such spinors can be viewed as a matrix $M^{\alpha\beta}$. The Dirac matrices and products thereof form a complete basis to expand any matrix. There are 16 basis elements: five γ_μ , ten $\gamma_{\mu\nu}$, and the unit element I (the matrices $\gamma_{\mu\nu\rho}$ and $\gamma_{\mu\nu\rho\sigma}$ and $\gamma_1 \dots \gamma_5$ are linearly dependent on the set of 16 matrices. For example, $\gamma_1 \dots \gamma_5$ is proportional to the unit matrix because $\gamma_5 = \gamma_1 \gamma_2 \gamma_3 \gamma_4$ and $\gamma_{\mu\nu\rho\sigma} = \epsilon_{\mu\nu\rho\sigma\tau} \gamma^\tau$, while $\gamma_{\mu\nu\rho} = -\frac{1}{2} \epsilon_{\mu\nu\rho\sigma\tau} \gamma^{\sigma\tau}$). There is one subtlety: the Dirac matrices have one index up and one index down: $(\gamma_\mu)^\alpha_\beta$ (necessary, otherwise one could not multiply two Dirac matrices). But in (27.37) the index β is up on the right hand side, so we must raise the index β ; this we do with the (inverse of) the charge conjugation matrix. For $SO(2N + 1)$ it is unique (up to similarity transformations), and $C = C_-$ in $d = 3, 7$ while $C = C_+$ in $d = 5, 9$. Its index structure is as follows

$$C_{\alpha\beta} \quad \text{and} \quad (C^{-1})^{\beta\gamma} \quad (27.38)$$

Then the spinor indices in $C_\pm \gamma_\mu C_\pm^{-1} = \pm \gamma_\mu^T$ match

$$C_{\pm, \alpha\beta} (\gamma_\mu)^\beta_\gamma C_\pm^{-1, \gamma\delta} = \pm (\gamma^T)_\alpha^\delta \quad (27.39)$$

So we expand

$$\psi^\alpha \chi^\beta = c_0 C^{-1, \alpha\beta} + c_{1, \mu} (\gamma^\mu C^{-1})^{\alpha\beta} + c_{2, \mu\nu} (\gamma^{\mu\nu} C^{-1})^{\alpha\beta} \quad (27.40)$$

Since the c_k depend on the μ, ν indices, we have made that dependence explicit to avoid confusion. To find the coefficients c_k , we multiply by the inverse of the matrices and take the trace. First we multiply with $C_{\beta\alpha}$

$$\begin{aligned} C_{\beta\alpha} \psi^\alpha \chi^\beta &= (\psi^\alpha C_{\beta\alpha} \chi^\beta) = -\chi^\beta C_{\beta\alpha} \psi^\alpha = -\bar{\chi} \psi \\ &= c_0 \text{tr}(C C^{-1}) + c_1 \text{tr}(C \gamma^\mu C^{-1}) + c_2 \text{tr}(C \gamma^{\mu\nu} C^{-1}) = c_0 \times 4 \end{aligned} \quad (27.41)$$

We used that $\text{tr}(C \gamma^\mu C^{-1}) = \text{tr} \gamma^\mu = 0$, and idem $\text{tr} \gamma^{\mu\nu} = 0$. So

$$c_0 = -\frac{1}{4} (\chi^\beta C_{\beta\alpha} \psi^\alpha). \quad (27.42)$$

The minus sign is produced if ψ^α and χ^β are anticommuting. (Sometimes one uses spinors which are commuting, then no minus sign is present.) Similarly one finds

$$(C \gamma_\nu)_{\beta\alpha} \psi^\alpha \chi^\beta = -(\chi^T C \gamma_\nu \psi) = c_{1, \mu} \text{tr}(C \gamma_\nu \gamma^\mu C^{-1}) = 4c_{1, \nu} \quad (27.43)$$

Finally

$$(C \gamma_{\mu\nu})_{\beta\alpha} \psi^\alpha \chi^\beta = -(\chi^T C \gamma_{\mu\nu} \psi) = c_{2, \rho\sigma} \text{tr}(C \gamma_{\mu\nu} \gamma^{\rho\sigma} C^{-1}) = c_{2, \rho\sigma} \text{tr}(\gamma_{\mu\nu} \gamma^{\rho\sigma}) \quad (27.44)$$

The trace $\text{tr}(\gamma_{\mu\nu} \gamma^{\rho\sigma})$ is only then nonvanishing if $\mu = \rho, \nu = \sigma$ or $\mu = \sigma, \nu = \rho$. If $\mu = \rho, \nu = \sigma$ we get $\gamma_{\mu\nu} \gamma^{\mu\nu}$ (no sum) $= -I$. Hence

$$c_{2, \rho\sigma} \text{tr}(\gamma_{\mu\nu} \gamma^{\rho\sigma}) = 4 \times (-2) c_{2, \mu\nu} \quad (27.45)$$

All in all we get

$$\begin{aligned} SO(5) : \quad \times \otimes \times &= \bullet \quad \oplus \quad \square \quad \oplus \quad \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \\ \psi^\alpha \chi^\beta &= -\frac{1}{4} \left[(C^{-1})^{\alpha\beta} (\chi^T C \psi) + (\gamma^\mu C^{-1})^{\alpha\beta} (\chi^T C \gamma_\mu \psi) - \frac{1}{2} (\gamma^{\mu\nu} C^{-1})^{\alpha\beta} (\chi^T C \gamma_{\mu\nu} \psi) \right] \\ \mathbf{4} \times \mathbf{4} &= \mathbf{1} \quad + \quad \mathbf{5} \quad + \quad \mathbf{10} \end{aligned} \quad (27.46)$$

Often one writes $\bar{\chi}$ for $\chi^T C$ where $\bar{\chi}$ is the Majorana conjugate of χ . Although we evaluated $\times \otimes \times$ for $SO(5)$, it can be done for any $SO(2N+1)$ in this way.

Next we consider in $SO(6)$ the Clebsch-Gordan expansion of two chiral spinors, or one chiral and one antichiral spinor. (We call a spinor in the s -rep a chiral spinor, and a spinor in the c -rep an antichiral spinor. That follows from the fact that the chirality matrix γ_c has the form $\gamma_c = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}$ in the representation of the Dirac matrices that we use.)

First consider $\psi^\alpha \chi^\beta$ where $\alpha, \beta = 1, \dots, 4$ are indices of the s -rep of $SO(6)$. We can again

expand on a complete basis of Dirac matrices. An arbitrary (nonchiral) spinor in 6 dimensions has 8 components, so the Dirac matrices (and products thereof) are 8×8 matrices which form the following basis

$$I, \gamma^\mu, \gamma^{\mu\nu}, \gamma^{\mu\nu\rho}, \gamma^{\mu\nu\rho\sigma}, \gamma^{\mu\nu\rho\sigma\tau}, \gamma_c \quad (27.47)$$

$$8 \times 8 = 64 = 1 + 6 + 15 + 20 + 15 + 6 + 1.$$

However, the spinors ψ^α and χ^β are chiral, and as a result only half this set of matrices will contribute because the Dirac matrices are block-offdiagonal (see (27.50)).

$$\begin{pmatrix} * & 0 \\ * & 0 \end{pmatrix}_\chi \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix}_{\text{one } \gamma \text{ matrix}} \begin{pmatrix} * \\ 0 \end{pmatrix}_\psi = 0; \quad \begin{pmatrix} * & 0 \\ * & 0 \end{pmatrix}_\chi \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix}_{\text{two } \gamma \text{ matrices}} \begin{pmatrix} * \\ 0 \end{pmatrix}_\psi \neq 0 \quad (27.48)$$

One might conclude that only the set with an even number of Dirac matrices contributes, but there is a subtlety: we did not yet include the charge conjugation matrix. So the issue is: which of the following expressions is nonvanishing

$$\begin{pmatrix} * & 0 \\ * & 0 \end{pmatrix} C \begin{pmatrix} * \\ 0 \end{pmatrix} \quad \text{OR} \quad \begin{pmatrix} * & 0 \\ * & 0 \end{pmatrix} C \gamma_\mu \begin{pmatrix} * \\ 0 \end{pmatrix} \quad (27.49)$$

It clearly depends on C : if C is block-diagonal then an even number of Dirac matrices contribute to the expression, but if C is block-offdiagonal, the terms with an odd number of Dirac matrices contribute. The block-structure of C depends on the dimensions; we consider the case of 6 dimensions.

So we must first find out whether C_+ and/or C_- are block-diagonal or block-offdiagonal in 6 dimensions. There might now seem to be 3 ways to solve this question

- a) Use the fact (often mentioned in class) that in Euclidean $d = 7$ a completely imaginary set of Dirac matrices exists. Then drop one of them, and one ends up with a set of Dirac matrices in $d = 6$. In even d , the irrep of Dirac matrices is unique. However, the 6-dimensional matrices thus obtained would not be block-offdiagonal (unless we picked them in 7 dimensions in a careful way). We could try to find a similarity transformation which casts them in block form but that is tedious.
- b) Use any representation of the Dirac matrices in $d = 5 + 0$, and tensor with Pauli matrices to get $d = 6 + 0$ Dirac matrices:

$$\gamma^{(6)} = \gamma^{(5)} \otimes \sigma_2; \quad I \otimes \sigma_1 \quad (27.50)$$

Then we can construct $C_\pm^{(6)}$ from $C_\pm^{(5)}$.

- c) Use the expression for Dirac matrices as a direct product of (in this case) three two-dimensional spaces.

We use b). As charge conjugation matrices in $d = 6$ we assume a direct product

$$C^{(6)} = C^{(5)} \otimes B^{(2)} \quad (27.51)$$

where $B^{(2)}$ are matrices to be determined (they will be related to the charge conjugation matrices $C_+^{(2)}$ and $C_-^{(2)}$ in $d = 2 + 0$). The $C^{(5)}$ satisfy $C^{(5)}\gamma^\mu C^{(5),-1} = +\gamma^{\mu,T}$ with $\mu = 1, \dots, 5$. So in $d = 5 + 0$, $C = C_+$. We then get two cases to study

$$C_+^{(6)} = C^{(5)} \otimes B_+^{(2)} \quad \text{and} \quad C_-^{(6)} = C^{(5)} \otimes B_-^{(2)} \quad (27.52)$$

In each case we get two conditions on $B^{(2)}$.

$$1) \ C_\pm^{(6)}(\gamma^\mu \otimes \sigma_2)C_\pm^{(6),-1} = C^{(5)}\gamma^\mu C^{(5),-1} \otimes B_\pm^{(2)}\sigma_2 B_\pm^{(2),-1} = \pm(\gamma^\mu \otimes \sigma_2)^T = \pm\gamma^{\mu,T} \otimes (-\sigma_2).$$

$$2) \ C_\pm^{(6)}(I \otimes \sigma_1)C_\pm^{(6),-1} = C^{(5)}IC^{(5),-1} \otimes B_\pm^{(2)}\sigma_1 B_\pm^{(2),-1} = \pm(I \otimes \sigma_1)^T = \pm I \otimes \sigma_1.$$

So we get the following conditions on $B_\pm^{(2)}$:

$$C_+^{(6)} = C^{(5)} \otimes B_+^{(2)} : \begin{array}{l} B_+^{(2)}\sigma_2 B_+^{(2),-1} = -\sigma_2 \\ B_+^{(2)}\sigma_1 B_+^{(2),-1} = \sigma_1 \end{array} ; \quad C_-^{(6)} = C^{(5)} \otimes B_-^{(2)} : \begin{array}{l} B_-^{(2)}\sigma_2 B_-^{(2),-1} = \sigma_2 \\ B_-^{(2)}\sigma_1 B_-^{(2),-1} = -\sigma_1 \end{array} \quad (27.53)$$

Clearly $B_+^{(2)} = \sigma_1$ and $B_-^{(2)} = -i\sigma_2$. (The $-i$ is added to make $B_-^{(2)}$ real but that is not essential.) Both are block-offdiagonal (as expected, $C_- = C_+\Gamma_{\text{chiral}}$ and $\Gamma_{\text{chiral}} = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} = I \otimes \sigma_3$ is block-diagonal).

$$C_+^{(6)} = \begin{pmatrix} 0 & C^{(5)} \\ C^{(5)} & 0 \end{pmatrix}; \quad C_-^{(6)} = \begin{pmatrix} 0 & -C^{(5)} \\ C^{(5)} & 0 \end{pmatrix} \quad (27.54)$$

(In odd dimensions C has both indices down, as $C_{\alpha\beta}$, but in even dimensions we write a spinor as $\Psi = \begin{pmatrix} \psi^\alpha \\ \bar{\chi}_\alpha \end{pmatrix}$ and then C has a more complicated index structure which depends on whether it is block-diagonal or block-offdiagonal. We discuss this below, but for the time-being we ignore this subtlety and write indices as indices for nonchiral spinors in the same way as in odd dimensions.)

Now we get

$$\times \otimes \times = \psi^\alpha \chi^\beta = \sum_{k=\text{odd}} a_k (\gamma^{\mu_1 \dots \mu_k} C^{-1})^{\alpha\beta} (\chi^T C \gamma_{\mu_1 \dots \mu_k} \psi) \quad (27.55)$$

with an odd number of Dirac matrices: γ^μ and $\gamma^{\mu\nu\rho}$, 6 + 10 matrices.

$$\begin{array}{llll} SO(6) : \quad \times \otimes \times = & \square & \oplus & \begin{array}{c} \square \\ \square \\ \square \end{array} \\ \psi^\alpha \chi^\beta = & -\frac{1}{4}(\gamma^\mu C^{-1})^{\alpha\beta} (\chi^T C \gamma_\mu \psi) & + & \frac{1}{24}(\gamma^{\mu\nu\rho} C^{-1})^{\alpha\beta} (\chi^T C \gamma_{\mu\nu\rho} \psi) \\ \mathbf{4} \times \mathbf{4} = & \mathbf{6} & + & \mathbf{10} \end{array} \quad (27.56)$$

We used that for chiral spinors the contribution from $\gamma_{\mu_1 \dots \mu_k}$ is the same as that from $\gamma_{\mu_1 \dots \mu_{6-k}}$, so we evaluated the contributions from γ_μ and $\gamma_{\mu\nu\rho}$ and multiplied the result by a factor 2. (There are

20 matrices $\gamma^{\mu\nu\rho}$ but in $d = 6$ only half of them are independent if they act on chiral or antichiral spinors. For example γ^{123} gives the same result as γ^{456} .) This agrees with the fact that the s -irrep of $SO(6)$ corresponds to the defining rep **4** of $SU(4)$, and in $SU(4)$ we have

$$SU(4) : \quad \mathbf{4} \times \mathbf{4} = \square \otimes \square = \square\square \oplus \begin{array}{|c|} \hline \square \\ \hline \end{array} = \mathbf{10} + \mathbf{6}. \quad (27.57)$$

Finally $\times \otimes +$. Now we get

$$\psi^\alpha \chi_{\dot{\beta}} = \sum_{k=\text{even}} a_k (\gamma^{\mu_1 \dots \mu_k} C^{-1})^{\alpha\beta} (\chi^T C \gamma_{\mu_1 \dots \mu_k} \psi) \quad (27.58)$$

with an even number of Dirac matrices: I and $\gamma^{\mu\nu}$, $1 + 15$ matrices. So for $SO(6)$ we get

$$\begin{aligned} SO(6) : \quad \times \otimes + &= \bullet \oplus \begin{array}{|c|} \hline \square \\ \hline \end{array} \\ \psi^\alpha \chi_{\dot{\beta}} &= -\frac{1}{4} (C^{-1})^\alpha_{\dot{\beta}} (\chi^T C \psi) + \frac{1}{8} (\gamma^{\mu\nu} C^{-1})^\alpha_{\dot{\beta}} (\chi^T C \gamma_{\mu\nu} \psi) \\ \mathbf{4}_s \times \mathbf{4}_c &= \mathbf{1} + \mathbf{15} \end{aligned} \quad (27.59)$$

Again the contributions from $\gamma_{\mu\nu\rho\sigma}$ and γ_{123456} are the same as those of $\gamma_{\mu\nu}$ and I , so we evaluated only the latter and multiplied the result by 2. In $SU(4)$ the same CG expansion yields

$$SU(4) : \quad \mathbf{4} \times \mathbf{4}^* = \square \otimes \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} = \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} = \mathbf{1} + \mathbf{15}. \quad (27.60)$$

This confirms also the expansion of $\times \otimes +$.

Let us now come back to the issue of the indices of $C^{(6)}$ and the Dirac matrices. We are led to the analogue of what are called in $d = 4$ two-component spinors, or dotted and undotted spinors, or van der Waerden formalism. Namely, when we decompose an 8-component nonchiral spinor Ψ into a chiral part of ψ^α and an antichiral part $\chi_{\dot{\alpha}}$ as

$$\Psi^A = \begin{pmatrix} \psi^\alpha \\ \chi_{\dot{\alpha}} \end{pmatrix} \quad \begin{array}{l} \alpha = 1, \dots, 4 \\ \dot{\alpha} = 1, \dots, 4 \end{array} \quad (27.61)$$

then the structure of the Dirac matrices Γ_M in $d = 6$ given in (27.50) is as follows

$$\Gamma_M = \begin{pmatrix} 0 & -i\sigma_M^{\alpha\dot{\beta}} \\ i\bar{\sigma}_{M,\dot{\alpha}\beta} & 0 \end{pmatrix}_{8 \times 8} \quad \begin{array}{l} \sigma_M = \{\sigma_i, \sigma_6\} = \{\gamma_i^{(5)}, iI\} \\ \bar{\sigma}_M = \{\bar{\sigma}_i, \bar{\sigma}_6\} = \{\gamma_i^{(5)}, -iI\} \end{array} \quad (27.62)$$

where $M = 1, \dots, 6$, $\alpha = 1, \dots, 4$ and $\dot{\beta} = 1, \dots, 4$. This structure is familiar from $d = 4$, but it holds in any even dimension. In particular, the matrix σ_M are often called generalized Pauli matrices. Then the indices of the charge conjugation matrices are fixed by the relation $C_\pm^{(6)} \Gamma_M = \pm \Gamma_M^T C_\pm^{(6)}$, or, easier, by consistency of the indices in the CG decomposition in (27.56)

and (27.59).

$$C_{\pm}^{(6)} = \begin{pmatrix} 0 & C_{\alpha}^{(5)\dot{\beta}} \\ \pm C^{(5)\dot{\alpha}}_{\beta} & 0 \end{pmatrix} \quad (27.63)$$

The matrices $C_{\alpha}^{(5)\dot{\beta}}$ and $C^{(5)\dot{\alpha}}_{\beta}$ are both numerically equal to the charge conjugation matrix $C^{(5)}$ in $d = 5$; only the indices have been written up or down to achieve consistency in $d = 6$.

Finally we note that in the CG decomposition it makes no difference whether one uses $C_+^{(6)}$ or $C_-^{(6)}$ since the minus sign in $C_-^{(6)}$ cancels (C_- appears in two places, in terms such as $(\gamma^{\mu}C^{-1})^{\alpha\beta}$ and in terms such as $\psi^T C \gamma_{\mu} \psi$). In fact, one can rewrite the CG decomposition given above in “two-component” spinor notation, using “Pauli” matrices instead of Dirac matrices. Then one really needs to put the dotted and undotted indices in the correct positions.

27.2 Spinor calculus and invariant spinor-tensors

The Clebsch-Gordan decompositions of the product of two spinor irreps, for example the relation

$$\begin{aligned} SO(5) : \times \otimes \times = & \quad \mathbf{1} \quad + \quad \mathbf{5} \quad + \quad \mathbf{10} \\ \psi^{\alpha} \otimes \chi^{\beta} = & -\frac{1}{4} \left[(C^{-1})^{\alpha\beta} (\chi^T C \psi) + (\gamma^{\mu} C^{-1})^{\alpha\beta} (\chi^T C \gamma_{\mu} \psi) - \frac{1}{2} (\gamma^{\mu\nu} C^{-1})^{\alpha\beta} (\chi^T C \gamma_{\mu\nu} \psi) \right] \end{aligned} \quad (27.64)$$

are $SO(5)$ -covariant by which is meant that the left-hand side and the right-hand side transform the same way under $SO(5)$ transformations of ψ and χ . But if one transforms ψ and χ , the left-hand side clearly transforms as a bispinor $\phi^{\alpha\beta}$, whereas on the right-hand side the transformation of ψ and χ does not seem to be that of a bispinor. Actually, the right-hand side also transforms as a bispinor because

- (i) transforming ψ and χ in $\chi^T C \psi$ and $\chi^T C \gamma_{\mu} \psi$ and $\chi^T C \gamma_{\mu\nu} \psi$, these expressions transform as a scalar, a vector, and a tensor, respectively
- (ii) transforming the index μ of $(\gamma^{\mu} C^{-1})^{\alpha\beta}$ as a vector, this expression transforms as a bispinor because $(\gamma^{\mu} C^{-1})^{\alpha\beta}$ is an invariant tensor as we now discuss. Similarly, transforming the indices μ, ν of $(\gamma^{\mu\nu} C^{-1})^{\alpha\beta}$ as a tensor, the expression $(\gamma^{\mu\nu} C^{-1})^{\alpha\beta}$ transforms as a bispinor. And finally, transforming $(C^{-1})^{\alpha\beta}$ as a bispinor, it does not change: $(C^{-1})^{\alpha\beta}$ is also an invariant tensor. Thus the decomposition in (27.64) is covariant.

Actually, for $SO(2N + 1)$ the following are invariant spinor-tensors: $(\gamma^{\mu})^{\alpha}_{\beta}$, $C_{\alpha\beta}$, $(C^{-1})^{\alpha\beta}$ and products thereof. We shall first prove this, and then analyze the case of $SO(2N)$ where the generalized Pauli matrices $(\sigma_M)^{\alpha\dot{\beta}}$, $\bar{\sigma}_{M,\dot{\alpha}\beta}$ and the charge conjugation matrices C_+ and C_- are also invariant tensors.

$SO(2N + 1)$. Consider in $2N + 1$ dimensions the Dirac matrices $(\gamma^{\mu})^{\alpha}_{\beta}$. They must have one spinor index up and the other spinor index down in order that one can multiply two Dirac matrices

$((\gamma^\mu)^\alpha_\beta (\gamma^\nu)^\beta_\gamma = (\gamma^\mu \gamma^\nu)^\alpha_\gamma)$. A spinor χ^α has the spinor index α up in order that γ^μ can act on χ $((\gamma^\mu)^\alpha_\beta \chi^\beta)$, and it transforms as follows under $SO(2N+1)$ transformations

$$\delta\chi^\alpha = \frac{1}{2}\lambda^{mn}\frac{1}{2}(\gamma_{mn})^\alpha_\beta\chi^\beta = \frac{1}{4}\lambda^{mn}(\gamma_{mn})^\alpha_b\chi^b \quad (27.65)$$

where $\lambda^{mn} = -\lambda^{nm}$ and $\gamma_{mn} = \frac{1}{2}(\gamma_m\gamma_n - \gamma_n\gamma_m)$ because, as we have shown, $M_{mn}^{spin} = \frac{1}{2}\gamma_{mn}$ satisfy the $SO(2N+1)$ Lie algebra. The factor $\frac{1}{2}$ in front of (27.65) is needed to avoid double-counting the cases $m > n$ and $m < n$. One can define a spinor with an index down by using the (unique in odd dimensions) charge conjugation matrix $C_{\alpha\beta}$ as follows

$$\begin{aligned} \chi^\beta C_{\beta\alpha} &= \chi_\alpha = C_{\alpha\beta}^T \chi^\beta = \sigma C_{\alpha\beta} \chi^\beta \\ C^T &= \sigma C \quad (C_{\beta\alpha} = \sigma C_{\alpha\beta}) \quad \sigma = \pm 1 \end{aligned} \quad (27.66)$$

The reason of this definition of χ_α (instead of $\chi_\alpha = C_{\alpha\beta}\chi^\beta$) is our wish to agree with standard conventions in $d=5$ (and later in $d=4$). We claim that χ_α transforms as follows

$$\delta\chi_\alpha = -\frac{1}{4}\chi_\gamma\lambda^{mn}(\gamma_{mn})^\gamma_\alpha \quad (27.67)$$

which implies that $\chi_\alpha\psi^\alpha$ is invariant. To derive this result for $\delta\chi_\alpha$ from $\delta\chi^\alpha$ in (27.65) we need

$$C_\pm\gamma^\mu C_\pm^{-1} = \pm\gamma^{\mu,T} \text{ or } C_{\pm,\alpha\beta}(\gamma^\mu)^\beta_\gamma = \pm(\gamma^{\mu,T})_\alpha^\delta (C_\pm)_{\delta\gamma} \quad (27.68)$$

(with either the $+$ or the $-$ spin in odd dimensions, not both) as well as $C^T = \sigma C$ where $\sigma = +1$ or -1 depending on the dimension $2N+1$. Then we get

$$\begin{aligned} \delta\psi_\alpha &= \sigma C_{\alpha\beta} \frac{1}{4}\lambda^{mn}(\gamma_{mn})^\beta_\gamma \psi^\gamma \quad (\text{transpose}) \\ &= \sigma \frac{1}{4}\lambda^{mn} \psi^\gamma (\gamma_n^T)_{\gamma}{}^\beta (\gamma_m^T)_{\beta}{}^\delta C_{\delta\alpha}^T \quad (\text{use } C^T = \sigma C) \\ &= \frac{1}{4}\lambda^{mn} \psi^\gamma C_{\gamma\delta} (\gamma_n)^\delta_\beta (\gamma_m)^\beta_\alpha \quad (\text{twice a } - \text{ sign}) \\ &= \frac{1}{4}\lambda^{mn} (\sigma C_{\delta\gamma} \psi^\gamma) (\gamma_{nm})^\delta_\alpha \quad (\text{since } \sigma C = C^T) \\ &= -\frac{1}{4}\lambda^{mn} \psi_\delta (\gamma_{mn})^\delta_\alpha \quad (\text{since } \gamma_{nm} = -\gamma_{mn}) \end{aligned} \quad (27.69)$$

We are now ready to prove that $(\gamma^\mu)^\alpha_\beta$ is one invariant tensor. We use the rules for $\delta\chi^\alpha$ and $\delta\psi_\alpha$ to transform the spinor indices, and transfer the index μ as a vector ($\delta v^\mu = \lambda^\mu_\nu v^\nu$)

$$\begin{aligned} \delta(\gamma^\mu)^\alpha_\beta &= \frac{1}{4}\lambda^{mn}(\gamma_{mn})^\alpha_{\alpha'} (\gamma^\mu)^{\alpha'}_\beta - \frac{1}{4}\lambda^{mn} \gamma^{\mu,\alpha}_{\beta'} (\gamma_{mn})^{\beta'}_\beta + \lambda^\mu_\nu (\gamma^\nu)^a_\beta \\ &= \frac{1}{4}\lambda^{mn} [\gamma_{mn}, \gamma^\mu]^\alpha_\beta + \lambda^\mu_\nu (\gamma^\nu)^a_\beta \end{aligned} \quad (27.70)$$

Using¹ $[\gamma_{mn}, \gamma^\mu] = 2\delta_n^\mu \gamma_m - 2\delta_m^\mu \gamma_n$ we obtain the desired result

$$\delta(\gamma^\mu)^\alpha{}_\beta = \frac{1}{4}\lambda^{mn}(4\delta_n^\mu \gamma_m)^\alpha{}_\beta + \lambda^\mu{}_\nu(\gamma^\nu)^\alpha{}_\beta = 0. \quad (27.71)$$

It is also straightforward to show that the charge conjugation matrix $C_{\alpha\beta}$, (and its inverse $C^{-1,\alpha\beta}$) are invariant tensors. From (27.67) it follows that we have to show that the following expression vanishes

$$\delta C_{\alpha\beta} = C_{\alpha'\beta}(-\frac{1}{4}\lambda^{mn})(\gamma_{mn})^{\alpha'}{}_\alpha + C_{\alpha\beta'}(-\frac{1}{4}\lambda^{mn})(\gamma_{mn})^{\beta'}{}_\beta \quad (27.72)$$

In matrix notation we get the combination $C\gamma_{mn} + (\gamma_{mn})^T C$, and the combination $C\gamma_{mn}C^{-1} + (\gamma_{mn})^T$ vanishes since $C\gamma_{mn}C^{-1} = \gamma_{[m}^T \gamma_{n]}^T = (\gamma_{nm})^T = -(\gamma_{mn})^T$. Thus $C_{\alpha\beta}$ is an invariant tensor. Also $\delta_\alpha{}^\beta$ is an invariant tensor. Products of invariant tensors are invariant tensors, so the matrices $C^{-1,\alpha\beta}$, $(\gamma^\mu C^{-1})^{\alpha\beta}$ and $(\gamma^{\mu\nu} C^{-1})^{\alpha\beta}$ in (27.64) are also invariant tensors.

Let us now demonstrate the covariance of (27.64) in detail. Transforming χ and ψ in $\chi^T C \gamma_\mu \psi$ as

$$\delta\chi_\alpha = -\chi_\beta \frac{1}{4}\lambda^{mn}(\gamma_{mn})^\beta{}_\alpha \quad \text{and} \quad \delta\psi^\beta = \frac{1}{4}\lambda^{mn}(\gamma_{mn})^\beta{}_\delta \psi^\delta \quad (27.73)$$

one obtains

$$\begin{aligned} \delta(\chi^T C \gamma_\mu \psi) &= (\delta\chi_\alpha)(\gamma^\mu)^\alpha{}_\beta \psi^\beta + \chi_\alpha (\gamma^\mu)^\alpha{}_\beta \delta\psi^\beta \\ &= \chi_\alpha [\gamma_\mu, \frac{1}{4}\lambda^{mn}\gamma_{mn}]^\alpha{}_\beta \psi^\beta = \lambda_\mu{}^n \chi^T C \gamma_n \psi. \end{aligned} \quad (27.74)$$

Acting with the matrix $\lambda_\mu{}^n$ on the factor $(\gamma^\mu C^{-1})^{\alpha\beta}$ in (27.64) we obtain

$$-\lambda_\mu{}^n (\gamma^\mu C^{-1})^{\alpha\beta}, \quad (27.75)$$

and using that transforming all three indices μ, α, β of $(\gamma^\mu C^{-1})^{\alpha\beta}$ leaves it invariant, we see that $-\lambda_\mu{}^n (\gamma^\mu C^{-1})^{\alpha\beta}$ is equal to the transformation of a bispinor $\phi^{\alpha\beta}$. The other two terms are handled the same way. Thus (27.64) is indeed a covariant equation.

In 4 dimensions one usually follows conventions in which $\psi^\alpha \chi_\alpha = -\psi_\alpha \chi^\alpha$ with $\alpha = 1, 2$. In $2N + 1$ dimensions we find instead, $\chi_\alpha = \chi^\beta C_{\beta\alpha} = \sigma C_{\alpha\beta} \chi^\beta$,

$$\begin{aligned} \psi^\alpha \chi_\alpha &= \psi^\alpha \sigma C_{\alpha\beta} \chi^\beta = -\chi^\beta \psi^\alpha \sigma C_{\alpha\beta} \quad (\text{if } \psi \text{ and } \chi \text{ are anticommuting}) \\ &= -\sigma \chi^\beta \psi_\beta = \sigma \psi_\alpha \chi^\alpha \end{aligned} \quad (27.76)$$

Note that it follows from our definition $\chi^\beta C_{\beta\alpha} = \chi_\alpha$ that $\chi^\beta = \chi_\alpha C^{-1,\alpha\beta}$. We would like to follow uniformly a northwest-southeast contraction convention ($\chi^\alpha \psi_\alpha$). The law $\chi_\alpha = \chi^\beta C_{\beta\alpha}$ follows this convention, but for χ^β we would like to use a matrix $C^{\beta\alpha}$ such that $\chi^\beta = C^{\beta\alpha} \chi_\alpha$. The definition

¹Use $[\gamma_{mn}, \gamma^\mu] = [\gamma_m \gamma_n - \delta_{mn}, \gamma^\mu] = [\gamma_m \gamma_n, \gamma^\mu] = \gamma_m \{\gamma_n, \gamma^\mu\} - \{\gamma_m, \gamma^\mu\} \gamma_n$.

of the matrix $C^{\beta\alpha}$ follows from the equality of $\chi^\beta = \chi_\alpha C^{-1,\alpha\beta}$ and $\chi^\beta = C^{\beta\alpha} \chi_\alpha$

$$C^{\beta\alpha} = (C^{-1})^{\alpha\beta} = \sigma(C^{-1})^{\beta\alpha}. \quad (27.77)$$

With the northwest-southeast contraction convention to lower both indices of $C^{\alpha\beta}$ with $C_{\gamma\delta}$ we find $C_{\alpha\beta}$

$$C^{\alpha'\beta'} C_{\alpha'\alpha} C_{\beta'\beta} = \sigma C^{-1,\alpha'\beta'} C_{\alpha'\alpha} C_{\beta'\beta} = \sigma \delta_{\beta}^{\alpha'} C_{\alpha'\alpha} = \sigma C_{\beta\alpha} = C_{\alpha\beta}. \quad (27.78)$$

We shall exclusively use $C_{\alpha\beta}$ and $C^{\alpha\beta}$ to raise and lower indices in $d = 2N + 1$, but note that $C_{\alpha\beta} C^{\beta\gamma} = C_{\alpha\beta} \sigma C^{-1,\beta\gamma} = \sigma \delta_{\alpha}^{\gamma}$. A more illuminating way to write this result is $C_{\alpha\beta} C^{\gamma\beta} = \delta_{\alpha}^{\gamma}$, which parallels the relation $\epsilon_{\alpha\beta} \epsilon^{\gamma\beta} = \delta_{\alpha}^{\gamma}$ of four dimensions.

$SO(2N)$. For $SO(2N)$ the analysis is more subtle. An arbitrary nonchiral spinor ψ is decomposed into a spinors ψ^α and $\chi_{\dot{\beta}}$ as follows

$$\Psi = \begin{pmatrix} \psi^\alpha \\ \chi_{\dot{\beta}} \end{pmatrix} \quad \alpha, \dot{\beta} = 1, \dots, 2^{N-1}. \quad (27.79)$$

A useful representation of the Dirac matrices Γ_M is obtained by setting $\Gamma_m = \gamma_m \otimes \sigma_2$ and $\Gamma_{2N} = I \otimes \sigma_1$, where γ_m are Dirac matrices in $2N - 1$ dimensions. Since the product of all $2N - 1$ Dirac matrices γ_m is proportional to the unit matrix (recall that γ_{2N-1} is proportional to the product of all other Dirac matrices), the chirality matrix Γ_c in $2N$ dimensions is proportional to $I \otimes \sigma_3$, and we take it exactly equal, $\Gamma_c = I \otimes \sigma_3$. Then ψ^α is chiral and $\chi_{\dot{\beta}}$ is antichiral. The Dirac matrices can be expressed in terms of generalized Pauli matrices as follows

$$\begin{aligned} \Gamma_M &= \begin{pmatrix} 0 & -i\sigma_M^{\alpha\dot{\beta}} \\ i\bar{\sigma}_{M,\dot{\alpha}\beta} & 0 \end{pmatrix} \quad \begin{array}{l} \sigma_M = \{\gamma_m, iI\} \\ \bar{\sigma}_M = \{\gamma_m, -iI\} \end{array} \quad M = 1, \dots, 2N \\ \Gamma_m &= \begin{pmatrix} 0 & -i\gamma_m \\ i\gamma_m & 0 \end{pmatrix} \quad m = 1, \dots, 2N - 1; \quad \Gamma_{2N} = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix} \end{aligned} \quad (27.80)$$

The $SO(2N)$ generators $M_{MN}^{\text{spin}} = \frac{1}{2}\Gamma_{MN} = \frac{1}{2}\Gamma_{[M}\Gamma_{N]}$ are block-diagonal and have the following form

$$M_{mn} = \begin{pmatrix} \frac{1}{2}(\sigma_{mn})^{\alpha\beta} & 0 \\ 0 & \frac{1}{2}(\bar{\sigma}_{mn})_{\dot{\alpha}\dot{\beta}} \end{pmatrix}; \quad M_{m,2N} = \begin{pmatrix} -\frac{i}{2}(\sigma_m)^{\alpha\beta} & 0 \\ 0 & \frac{i}{2}(\bar{\sigma}_m)_{\dot{\alpha}\dot{\beta}} \end{pmatrix} \quad (27.81)$$

where $M, N = 1, \dots, 2N$ and $m, n = 1, \dots, 2N - 1$ while

$$\begin{aligned}
(\sigma_{mn})^\alpha{}_\beta &= \frac{1}{2} \left[(\sigma_m)^{\alpha\dot{\gamma}} \bar{\sigma}_{n,\dot{\gamma}\beta} - (\sigma_n)^{\alpha\dot{\gamma}} \bar{\sigma}_{m,\dot{\gamma}\beta} \right] \\
(\bar{\sigma}_{mn})_{\dot{\alpha}}{}^{\dot{\beta}} &= \frac{1}{2} \left[\bar{\sigma}_{m,\dot{\alpha}\gamma} (\sigma_n)^{\gamma\dot{\beta}} - \bar{\sigma}_{n,\dot{\alpha}\gamma} (\sigma_m)^{\gamma\dot{\beta}} \right] \\
-i(\sigma_m)^\alpha{}_\beta &= -i(\sigma_m)^{\alpha\dot{\gamma}} \delta_{\dot{\gamma}\beta} \\
i(\bar{\sigma}_m)_{\dot{\alpha}}{}^{\dot{\beta}} &= i\bar{\sigma}_{m,\dot{\alpha}\gamma} \delta^{\gamma\dot{\beta}}
\end{aligned} \tag{27.82}$$

All σ_m and $\bar{\sigma}_m$ are numerically equal to γ_m , but σ_m has indices $\sigma_m^{\alpha\dot{\beta}}$ while $\bar{\sigma}_m$ has indices $\sigma_{m,\dot{\alpha}\beta}$. The $\frac{1}{2}\sigma_{MN} = \frac{1}{2}(\sigma_{mn}, -i\sigma_m)$ and $\frac{1}{2}\bar{\sigma}_{MN} = \frac{1}{2}(\bar{\sigma}_{mn}, i\bar{\sigma}_m)$ satisfy the $SO(2N)$ Lie algebra (but note the position of the indices $\dot{\gamma}$ in σ_{mn} and γ in $\bar{\sigma}_{mn}$). The problem is now to show that $(\sigma_M)^{\alpha\dot{\beta}}$, $\bar{\sigma}_{M,\dot{\alpha}\beta}$, $(\sigma_{MN})^\alpha{}_\beta$ and $(\bar{\sigma}_{MN})_{\dot{\alpha}}{}^{\dot{\beta}}$ are all invariant spinor-tensors, as are the charge conjugation matrices C_+ and C_- . For the latter, however, we must be aware of a subtlety: they can be either block-diagonal or block-off-diagonal, and these blocks carry different indices in these two cases. There is a straightforward way to derive the results by starting with the corresponding results for the Dirac matrices and then decomposing into blocks. The latter are just as in $2N + 1$ dimensions, and by decomposing Γ_M as in (27.80) we can deduce the results for σ_M and $\bar{\sigma}_M$. Similarly from $C_\pm \Gamma_M C_\pm^{-1} = \pm \Gamma_M^T$ we can deduce corresponding results for the blocks which constitute C_+ and C_- . As a check we shall decompose the reducible spinor invariant $\Psi^T C_\pm X$ into chiral and antichiral parts, and check the invariance of the resulting two terms under $SO(2N)$ transformations. This checks that the charge conjugation matrices are invariant spinor-tensors. We repeat that all these results are deduced, there are no new definitions or ambiguities about the position of spinor indices. At the end we shall compare with the special case of $d = 3 + 1$ which is of course needed in much of $d = 3 + 1$ physics.

We shall express the charge conjugation matrices $C_\pm^{(2N)}$ in $d = 2N$ in terms of the (unique) charge conjugation matrix $C^{(2N-1)}$ in $d = 2N - 1$, and need the following results for the latter

$$\begin{aligned}
d = 3 : C &= C_- & C^T &= -C \\
d = 5 : C &= C_+ & C^T &= -C \\
d = 7 : C &= C_- & C^T &= C \\
d = 9 : C &= C_+ & C^T &= C \\
d = 11 : C &= C_- & C^T &= -C \\
&\vdots & &\vdots
\end{aligned} \tag{27.83}$$

These results are easily obtained by taking particular representations of the Dirac matrices in each dimension. For example, in $d = 7 + 0$ there exists a purely imaginary representation of the Dirac matrices, so $C = C_- = I$ in that case.

$d = 2N = 4n$. The charge conjugation matrices $C_{\pm}$ are block-diagonal and $C_+ = C_- \Gamma_c$

$$C_{\pm}^{(4n)} = \begin{pmatrix} C_{\alpha\beta}^{(4n-1)} & 0 \\ 0 & \pm C^{(4n-1),\dot{\alpha}\dot{\beta}} \end{pmatrix}; \quad \Psi = \begin{pmatrix} \psi^{\alpha} \\ \chi_{\dot{\alpha}} \end{pmatrix}; \quad X = \begin{pmatrix} \lambda^{\alpha} \\ \zeta_{\dot{\alpha}} \end{pmatrix} \quad (27.84)$$

Consider

$$\Psi^T C_{\pm}^{(4n)} X = \psi^{\alpha} C_{\alpha\beta}^{(4n-1)} \lambda^{\beta} \pm \chi_{\dot{\alpha}} C^{(4n-1),\dot{\alpha}\dot{\beta}} \zeta_{\dot{\beta}} \quad (27.85)$$

The position of indices on $C_{\pm}$ follow from the position of indices on the spinors. Invariance of the first of these two terms under $SO(2N)$ transformations requires

$$\begin{aligned} \psi^{\gamma} \frac{1}{4} \lambda^{MN} (\sigma_{MN}^T)_{\gamma}^{\alpha} C_{\alpha\beta}^{(4n-1)} \lambda^{\beta} + \psi^{\alpha} C_{\alpha\beta}^{(4n-1)} \frac{1}{4} \lambda^{MN} (\sigma_{MN})^{\beta}_{\gamma} \lambda^{\gamma} &= 0 \\ \text{or } C_{\alpha\beta}^{(4n-1)} (\sigma_{MN})^{\beta}_{\gamma} + (\sigma_{MN}^T)_{\alpha}^{\beta} C_{\beta\gamma}^{(4n-1)} &= 0 \end{aligned} \quad (27.86)$$

For $\sigma_{MN} = \sigma_{mn} = \gamma_{mn}$ this relation reduces to $C^{(4n-1)} \gamma_{mn} + \gamma_{mn}^T C^{(4n-1)}$ in $d = 4n - 1$ dimensions which always holds, whether $C^{(4n-1)} = C_+^{(4n-1)}$ or $C^{(4n-1)} = C_-^{(4n-1)}$. However, for $\sigma_{MN} = \sigma_{m,2N} = -i\gamma_m$, this relation reduces to

$$C_{\alpha\beta}^{(4n-1)} (-i\gamma_m)^{\beta}_{\gamma} + (-i\gamma_m^T)_{\alpha}^{\beta} C_{\beta\gamma}^{(4n-1)} = 0 \quad (27.87)$$

and this relation is only true since $C^{(4n-1)} = C_-^{(4n-1)}$ in $d = 3, 7$.

$d = 2N = 4n + 2$. Now the charge conjugation matrices are block off-diagonal

$$C_{\pm}^{(4n+2)} = \begin{pmatrix} 0 & \pm C_{\alpha}^{(4n+1)\dot{\beta}} \\ C^{(4n+1)\dot{\alpha}}_{\beta} & 0 \end{pmatrix} \quad (27.88)$$

and

$$\Psi^T C_{\pm}^{(4n+2)} X = \chi_{\dot{\alpha}} C^{(4n+1)\dot{\alpha}}_{\beta} \lambda^{\beta} \pm \psi^{\alpha} C_{\alpha}^{(4n+1)\dot{\beta}} \zeta_{\dot{\beta}} \quad (27.89)$$

The position of indices on $C_{\pm}$ follows from the position of indices on the spinors. The invariance of the first term under $SO(2N)$ transformation requires

$$\begin{aligned} \chi_{\dot{\beta}} \frac{1}{4} \lambda^{MN} (\bar{\sigma}_{MN}^T)^{\dot{\beta}}_{\dot{\alpha}} C^{(4n+1)\dot{\alpha}}_{\gamma} \lambda^{\gamma} + \chi_{\dot{\alpha}} C^{(4n+1)\dot{\alpha}}_{\beta} \frac{1}{4} \lambda^{MN} (\sigma_{MN})^{\beta}_{\gamma} \lambda^{\gamma} &= 0 \\ \text{or } C^{(4n+1)\dot{\alpha}}_{\beta} (\sigma_{MN})^{\beta}_{\gamma} + (\bar{\sigma}_{MN}^T)^{\dot{\alpha}}_{\dot{\beta}} C^{(4n+1)\dot{\beta}}_{\gamma} &= 0 \end{aligned} \quad (27.90)$$

Again we consider the two cases $\sigma_{MN} = \sigma_{mn} = \gamma_{mn}$ and $\sigma_{MN} = \sigma_{m,2N} = -i\gamma_m$ separately. In the former case we must prove the relation

$$C^{(4n+1)} \gamma_{mn} + \gamma_{mn}^T C^{(4n+1)} = 0 \quad (27.91)$$

which is again always true. But in the case $\sigma_{MN} = -i\gamma_m$ we get

$$C^{(4n+1)}(-i\gamma_m) + (i\gamma_m^T)C^{(4n+1)} = 0 \quad (27.92)$$

and this is only true since $C^{(4n+1)} = C_+^{(4n+1)}$.

Using these results it is now straightforward to check that $\sigma_M^{\alpha\dot{\beta}}$ and $\bar{\sigma}_{M,\dot{\alpha}\beta}$ are invariant spinor-tensors. Also the blocks into which the charge conjugation matrices $C_{\pm}^{(2N)}$ decompose, are invariant tensors. It follows that also the Clebsch-Gordan decompositions of products of spinor irreps of $SO(2N)$ such as in (27.56) and (27.59) are covariant equations under $SO(2N)$ transformations. Note that the matrices $C_{\alpha\beta}^{(4n)}$, $C^{(4n),\dot{\alpha}\dot{\beta}}$, $C^{(4n+2),\dot{\alpha}}_{\beta}$ and $C_{\alpha}^{(4n+2),\dot{\beta}}$ are numerically equal to the charge conjugation matrix on $(4n-1)$ and $(4n+1)$ dimensions, but they are invariant spinor-tensors of the full group $SO(4n)$ and $SO(4n+2)$, respectively.

We can now rewrite the Clebsch-Gordan decomposition of the product of two spinor irreps in terms of generalized Pauli matrices instead of Dirac matrices. For example, in $d=6$ (27.56) becomes

$$\begin{aligned} \psi^{\alpha}\chi^{\beta} = & -\frac{1}{4}\left(\sigma^{\mu}\alpha\dot{\beta}(C^{(5),-1})_{\dot{\beta}}^{\beta}\right)\left(\chi^{\alpha}C_{\alpha}^{(5),\dot{\beta}}\bar{\sigma}_{\mu,\dot{\beta}\gamma}\psi^{\gamma}\right) \\ & + \frac{1}{24}\left(\sigma^{[\mu,\alpha\dot{\beta}}\bar{\sigma}_{\dot{\beta}\gamma}^{\nu}\sigma^{\rho],\gamma\dot{\delta}}(C^{(5),-1})_{\dot{\delta}}^{\beta}\right)\left(\chi^{\alpha}C_{\alpha}^{(5),\dot{\beta}}\bar{\sigma}_{\mu,\dot{\beta}\gamma}\sigma_{\nu}^{\gamma\dot{\delta}}\bar{\sigma}_{\rho,\dot{\delta}\beta}\psi^{\beta}\right) \end{aligned} \quad (27.93)$$

or suppressing contracted indices

$$\begin{aligned} \times \otimes \times = & \quad \square \quad \oplus \quad \begin{array}{c} \square \\ \square \\ \square \end{array} \quad (27.94) \\ \psi^{\alpha}\chi^{\beta} = & -\frac{1}{4}(\sigma^{\mu}C^{(5),-1})^{\alpha\beta}(\chi^TC^{(5)}\bar{\sigma}_{\mu}\psi) + \frac{1}{24}(\sigma^{[\mu}\bar{\sigma}^{\nu}\sigma^{\rho]}C^{(5),-1})^{\alpha\beta}(\chi^TC^{(5)}\bar{\sigma}_{\mu}\sigma_{\nu}\bar{\sigma}_{\rho}\psi) \end{aligned}$$

Products of Dirac matrices such as $\gamma_{\mu}\gamma_{\nu}\gamma_{\rho}$ are replaced by $\sigma_{\mu}\bar{\sigma}_{\nu}\sigma_{\rho}$ or $\bar{\sigma}_{\mu}\sigma_{\nu}\bar{\sigma}_{\rho}$, depending on the spinor they act on. As even simpler notation is obtained by denoting $(\sigma^{\mu}C^{(5),-1})^{\alpha\beta}$ by $\sigma^{\mu,\alpha\beta}$, in which case one obtains

$$\psi^{\alpha}\chi^{\beta} = -\frac{1}{4}(\sigma^{\mu})^{\alpha\beta}(\chi^TC^{(5)}\bar{\sigma}_{\mu}\psi) + \frac{1}{24}(\sigma^{[\mu}\bar{\sigma}^{\nu}\sigma^{\rho]})^{\alpha\beta}(\chi^T\bar{\sigma}_{\mu}\sigma_{\nu}\bar{\sigma}_{\rho}\psi). \quad (27.95)$$

Chapter 28

Decomposing a tensor with Young symmetrizers

Given a Young tableau (really a Young pattern, see below) of, for example, $SU(N)$, we know how to compute the dimension of the irreducible representation (irrep) to which it corresponds (the “factors over hooks” formula). However, one sometimes wants to decompose a reducible tensor into irreps. The simplest case is an arbitrary tensor $T_{\mu\nu}$ with two indices; it can be decomposed into a symmetric part and an antisymmetric part, and both are irreps of $SU(N)$

$$T^{\mu\nu} = \frac{1}{2} (T^{\mu\nu} + T^{\nu\mu}) + \frac{1}{2} (T^{\mu\nu} - T^{\nu\mu}) . \quad (28.1)$$

This agrees of course with the rules for the Clebsch-Gordan decomposition of the product of two fundamental irreps

$$\square \otimes \square = \square\square \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \quad (28.2)$$

but (28.1) is more specific: it tells how the indices of the irreps are related to the indices of the reducible tensor $T^{\mu\nu}$. If one tries to do the same decomposition for an arbitrary tensor $T^{\mu\nu\rho}$ with three indices, one encounters already some problems. Clearly the totally symmetric part and the totally antisymmetric part are irreps, but in addition one expects to need some irreps with mixed symmetry, and then one encounters the problem of how many such mixed irreps occur. From the Clebsch-Gordan rules for $SU(N)$ one finds

$$\square \otimes \square \otimes \square = \left(\square\square \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \end{array} \right) \otimes \square = \square\square\square \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array} \oplus \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array} \quad (28.3)$$

and thus one expects two irreps with mixed symmetry, but the problem is how to write them with indices: to express the irreps as particular linear combinations of the components of $T^{\mu\nu\rho}$. We now describe how to achieve this for $SU(N)$. For other groups, one must further decompose the irreps of $SU(N)$ by taking ordinary traces for $SO(N)$ or symplectic traces for $USp(2N)$.

The rules for $SU(N)$ are easy to formulate. Consider a “**Young pattern**” which is a Young

tableau without numbers in it. For a k -index tensor $T_{\mu_1 \dots \mu_k}$ one needs all Young patterns with k boxes. For example, for $k = 3$, the Young patterns are

$$\begin{array}{|c|c|c|}, & \begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}, & \begin{array}{|c|} \hline \\ \hline \\ \hline \end{array} \end{array} \quad (28.4)$$

Next put the numbers $1, 2, \dots, k$ in the boxes in all ways such that

- the numbers increase in each row if one reads from left to right,
- the numbers also increase in each column if one reads from top to bottom.

Young patterns where the numbers are written down in any way, not necessarily satisfying these restrictions, are called **Young tableaux**, while if the restrictions are satisfied, they are called **standard Young tableaux**. For $k = 3$ and the Young pattern $\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}$ one has $6(= 3!)$ Young tableaux of which only two are standard Young tableaux.

$$\underbrace{\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array}}_{\text{Standard Young tableaux}} \quad \underbrace{\begin{array}{|c|c|} \hline 2 & 1 \\ \hline 3 & \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 2 & 3 \\ \hline 1 & \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 3 & 1 \\ \hline 2 & \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 3 & 2 \\ \hline 1 & \\ \hline \end{array}}_{\text{Young tableaux}} \quad (28.5)$$

Given a tensor $T_{\mu_1 \dots \mu_k}$ and a standard Young tableau,¹ one first symmetrizes the indices of $T_{\mu_1 \dots \mu_k}$ as indicated by each row, and then one antisymmetrizes the indices as indicated by each column. This will produce an irrep of $SU(N)$. For example, for $k = 3$ and the standard Young tableau $\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 & \\ \hline \end{array}$ one obtains the following irrep

$$\hat{T}_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}, I} = (T_{\mu_1 \mu_2 \mu_3} + T_{\mu_2 \mu_1 \mu_3}) - (T_{\mu_3 \mu_2 \mu_1} + T_{\mu_2 \mu_3 \mu_1}). \quad (28.6)$$

and from $\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 & \\ \hline \end{array}$ one finds the other irrep in (28.3) with mixed symmetry

$$\hat{T}_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}, II} = (T_{\mu_1 \mu_2 \mu_3} + T_{\mu_3 \mu_2 \mu_1}) - (T_{\mu_2 \mu_1 \mu_3} + T_{\mu_3 \mu_1 \mu_2}). \quad (28.7)$$

Each of these two irreps corresponds to the **8**, but they both are needed in the decomposition of $T_{\mu_1 \mu_2 \mu_3}$ into irreps. This decomposition reads as follows

$$T_{\mu_1 \mu_2 \mu_3} = \frac{1}{6} T_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array}} + \frac{1}{3} T_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}, I} + \frac{1}{3} T_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|c|} \hline & \\ \hline & \\ \hline \end{array}, II} + \frac{1}{6} T_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|} \hline \\ \hline \\ \hline \end{array}} \quad (28.8)$$

where

$$T_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|c|c|} \hline & & \\ \hline & & \\ \hline \end{array}} = T_{\mu_1 \mu_2 \mu_3} + T_{\mu_2 \mu_3 \mu_1} + T_{\mu_3 \mu_1 \mu_2} + T_{\mu_1 \mu_3 \mu_2} + T_{\mu_3 \mu_2 \mu_1} + T_{\mu_2 \mu_1 \mu_3} \quad (28.9)$$

and

$$T_{\mu_1 \mu_2 \mu_3}^{\begin{array}{|c|} \hline \\ \hline \\ \hline \end{array}} = T_{\mu_1 \mu_2 \mu_3} + T_{\mu_2 \mu_3 \mu_1} + T_{\mu_3 \mu_1 \mu_2} - T_{\mu_1 \mu_3 \mu_2} - T_{\mu_3 \mu_2 \mu_1} - T_{\mu_2 \mu_1 \mu_3}. \quad (28.10)$$

¹To simplify the notation, we write the indices on tensors as $T_{\mu_1 \dots \mu_k}$ instead of $T^{\mu_1 \dots \mu_k}$, but this is only a change in notation. The tensors $T_{\mu_1 \dots \mu_k}$ are still represented by Young tableaux, for example v_μ still corresponds to $\begin{array}{|c|} \hline \\ \hline \end{array}$.

It is straightforward to check this decomposition of $T_{\mu\nu\rho}$. The dimensions of the irreps are, of course, given by the factors over hooks rule:

$$N^3 = \frac{1}{6}N(N+1)(N+2) + 2 \times \frac{1}{3}N(N^2-1) + \frac{1}{6}N(N-1)(N-2) \quad (28.11)$$

Note that (28.7) cannot be obtained from (28.6) by interchanging the indices μ_2 and μ_3 in (28.6) because in both cases we start with the same term $T_{\mu_1\mu_2\mu_3}$.

One can define an inner product² for the components of $T_{\mu\nu\rho}$ as follows

$$(T_{\mu\nu\rho}, T_{\alpha\beta\gamma}) = \delta_{\mu\alpha}\delta_{\nu\beta}\delta_{\rho\gamma} \quad (28.12)$$

According to this inner product, $T^{\square\square\square}$ and $T^{\square}$ are orthonormal to all irreps, but $T^{\square I}$ and $T^{\square II}$ are not orthogonal to each other

$$(T^{\square I}, T^{\square II}) = -1 \quad (28.13)$$

However, one can take linear combination that are orthogonal, for example as follows

$$T^{\square I} + T^{\square II} = \beta T^{\square\square\square} + \left((1-\beta)T^{\square I} + T^{\square II}\right) \quad (28.14)$$

For $\beta = \frac{3}{4}$ the first and second tensor on the right-hand side are orthogonal. It is easier to use the basis on the left-hand side.

The theory behind these rules goes back to Young and Frobenius, and to the representation theory of the symmetric groups $S(n)$. The elements of $S(n)$ for given n can be written as products of cycles, such as (125)(37). Given a standard Young tableau T , one constructs the Young symmetrizer P (and the Young antisymmetrizer Q) as products of (anti)symmetrizers for each row (column), and the Young operator Y as $Y = cQP$, where c is a constant. (We shall presently discuss the tensors one gets if one uses PQ instead of QP .) Then YT is the irrep. For example

$$\begin{aligned} Y\left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 \\ \hline \end{array}\right) &= \frac{1}{3}(e - (13))(e + (12)) \quad \text{gives (28.6)} \\ Y\left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 \\ \hline \end{array}\right) &= \frac{1}{3}(e - (12))(e + (13)) \quad \text{gives (28.7)} \end{aligned} \quad (28.15)$$

The coefficients $\frac{1}{3}$ ensure that $Y^2 = Y$. One can also prove (and / or check) that the Y 's for different standard Young tableaux are orthogonal. For example

$$Y\left(\begin{array}{|c|c|} \hline 1 & 2 \\ \hline 3 \\ \hline \end{array}\right) Y\left(\begin{array}{|c|c|} \hline 1 & 3 \\ \hline 2 \\ \hline \end{array}\right) = \frac{1}{9}(e - (13))(e + (12))(e - (12))(e + (13)) = 0 \quad (28.16)$$

clearly vanishes because $(e + (12))(e - (12)) = 0$.

Let us now give the result for an arbitrary 4-index tensor $T_{\mu_1\mu_2\mu_3\mu_4}$. There are 10 standard

²An inner product (x, y) must satisfy four conditions: 1) positive definiteness: $(x, x) \geq 0$ and $(x, x) = 0$ iff $x = 0$, 2) linearity $(x, y + \alpha z) = (x, y) + \alpha(x, z)$ and 3) hermiticity $(x, y) = (y, x)^*$.

Young tableaux

$$\begin{array}{c} \boxed{1} \\ \boxed{2} \\ \boxed{3} \\ \boxed{4} \end{array}; \quad \begin{array}{cc} \boxed{1} & \boxed{2} \\ \boxed{3} & \\ \boxed{4} & \end{array}, \quad \begin{array}{cc} \boxed{1} & \boxed{3} \\ \boxed{2} & \\ \boxed{4} & \end{array}, \quad \begin{array}{cc} \boxed{1} & \boxed{4} \\ \boxed{2} & \\ \boxed{3} & \end{array}; \quad \begin{array}{cc} \boxed{1} & \boxed{2} \\ \boxed{3} & \boxed{4} \end{array}, \quad \begin{array}{cc} \boxed{1} & \boxed{3} \\ \boxed{2} & \boxed{4} \end{array}; \\
 \begin{array}{ccc} \boxed{1} & \boxed{2} & \boxed{3} \\ \boxed{4} & & \end{array}, \quad \begin{array}{ccc} \boxed{1} & \boxed{2} & \boxed{4} \\ \boxed{3} & & \end{array}, \quad \begin{array}{ccc} \boxed{1} & \boxed{3} & \boxed{4} \\ \boxed{2} & & \end{array}; \quad \begin{array}{cccc} \boxed{1} & \boxed{2} & \boxed{3} & \boxed{4} \end{array}
 \end{array} \quad (28.17)$$

So a general tensor $T_{\mu_1\mu_2\mu_3\mu_4}$ decomposes into 10 irreps of $SU(N)$. (For low N , there may be fewer irreps; for example, for $SU(3)$ the totally antisymmetric irrep $\begin{array}{c} \boxed{1} \\ \boxed{2} \\ \boxed{3} \end{array}$ vanishes.) The Clebsch-Gordan rules confirm that one gets 10 irreps

$$\begin{array}{c} \square \\ \square \\ \square \\ \square \end{array} \otimes \begin{array}{c} \square \\ \square \\ \square \\ \square \end{array} \otimes \begin{array}{c} \square \\ \square \\ \square \\ \square \end{array} \otimes \begin{array}{c} \square \\ \square \\ \square \\ \square \end{array} = \left(\begin{array}{cc} \square & \square \\ \square & \square \end{array} \oplus \begin{array}{c} \square \\ \square \end{array} \right) \otimes \begin{array}{c} \square \\ \square \\ \square \\ \square \end{array} = \\
 \left(\begin{array}{cccc} \square & \square & \square & \square \\ \square & \square & \square & \square \end{array} \oplus 2 \begin{array}{cc} \square & \square \\ \square & \square \end{array} \oplus \begin{array}{c} \square \\ \square \\ \square \end{array} \right) \otimes \begin{array}{c} \square \\ \square \\ \square \\ \square \end{array} = \begin{array}{cccc} \square & \square & \square & \square \\ \square & \square & \square & \square \end{array} \oplus 3 \begin{array}{ccc} \square & \square & \square \\ \square & & \end{array} \oplus 2 \begin{array}{cc} \square & \square \\ \square & \square \end{array} \oplus 3 \begin{array}{cc} \square & \square \\ \square & \square \end{array} \oplus \begin{array}{c} \square \\ \square \\ \square \end{array}. \quad (28.18)$$

To decompose $T_{\mu_1\mu_2\mu_3\mu_4}$ into irreps which are linear combinations of the components of $T_{\mu_1\mu_2\mu_3\mu_4}$, one may follow the simple rules which we gave at the beginning. This can be done, but the expressions are long. It is simpler to use Young operators. We then get the following decomposition into irreps

$$T_{\mu_1\mu_2\mu_3\mu_4} = (Y^{\begin{array}{c} \square \\ \square \\ \square \\ \square \end{array}} T)_{\mu_1\mu_2\mu_3\mu_4} + (Y^{\begin{array}{cc} \square & \square \\ \square & \square \end{array}, I} T)_{\mu_1\mu_2\mu_3\mu_4} + (Y^{\begin{array}{ccc} \square & \square & \square \\ \square & \square & \square \end{array}, II} T)_{\mu_1\mu_2\mu_3\mu_4} + \cdots + (Y^{\begin{array}{cccc} \square & \square & \square & \square \\ \square & \square & \square & \square \end{array}} T)_{\mu_1\mu_2\mu_3\mu_4}. \quad (28.19)$$

Some explicit formulas are

- 1) $(Y^{\begin{array}{cccc} \square & \square & \square & \square \\ \square & \square & \square & \square \end{array}} T) = \frac{1}{24}(e + (12) + \cdots + (4321))T$. There are 24 terms³ and this is the completely symmetric one-dimensional irrep. The factor $\frac{1}{24}$ is needed in order that $Y^2 = Y$.

$$\begin{aligned}
 2) \quad (Y^{\begin{array}{cc} \square & \square \\ \square & \square \end{array}, I} T)_{\mu_1\mu_2\mu_3\mu_4} &= \left(Y \left(\begin{array}{cc} \boxed{1} & \boxed{2} \\ \boxed{3} & \\ \boxed{4} & \end{array} \right) T \right)_{\mu_1\mu_2\mu_3\mu_4} \\
 &= \frac{1}{8}(e - (13) - (14) - (34) + (134) + (143))(e + (12))T_{\mu_1\mu_2\mu_3\mu_4} \\
 &= \frac{1}{8} \begin{pmatrix} e + (12) - (13) - (14) - (34) - (123) - (124) \\ -(12)(34) + (134) + (1234) + (143) + (1243) \end{pmatrix} T_{\mu_1\mu_2\mu_3\mu_4} \\
 &= \frac{1}{8} (T_{\mu_1\mu_2\mu_3\mu_4} + T_{\mu_2\mu_1\mu_3\mu_4} - T_{\mu_3\mu_2\mu_1\mu_4} + \cdots + T_{\mu_2\mu_4\mu_1\mu_3}). \quad (28.20)
 \end{aligned}$$

There are 12 terms in this expression. We used that $(123)T_{\mu_1\mu_2\mu_3\mu_4} = T_{\mu_2\mu_3\mu_1\mu_4}$ (“replace μ_1 by μ_2 , μ_2 by μ_3 and μ_3 by μ_1 in $T_{\mu_1\mu_2\mu_3\mu_4}$ ”) and we read the products of cycles from right to left.

³In expressions such as $e + (12)$, one adds group elements. In groups one can only multiply group elements, but in a “group algebra” one can both multiply and take linear combinations.

As one last example, we construct the irrep corresponding to $\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}$

$$\begin{aligned}
(Y^{\boxplus, I} T)_{\mu_1 \mu_2 \mu_3 \mu_4} &= \left(Y \left(\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix} \right) T \right)_{\mu_1 \mu_2 \mu_3 \mu_4} \\
&= \frac{1}{12} \left[(e - (13))(e - (24))(e + (12))(e + (34))T \right]_{\mu_1 \mu_2 \mu_3 \mu_4} \\
&= \frac{1}{12} \left[(e - (13))(e - (24)) \right] [T_{\mu_1 \mu_2 \mu_3 \mu_4} + T_{\mu_2 \mu_1 \mu_3 \mu_4} + T_{\mu_1 \mu_2 \mu_4 \mu_3} + T_{\mu_2 \mu_1 \mu_4 \mu_3}] \\
&= \frac{1}{12} \begin{bmatrix} T_{\mu_1 \mu_2 \mu_3 \mu_4} + T_{\mu_2 \mu_1 \mu_3 \mu_4} + T_{\mu_1 \mu_2 \mu_4 \mu_3} + T_{\mu_2 \mu_1 \mu_4 \mu_3} \\ -T_{\mu_3 \mu_2 \mu_1 \mu_4} - T_{\mu_2 \mu_3 \mu_1 \mu_4} - T_{\mu_3 \mu_2 \mu_4 \mu_1} - T_{\mu_2 \mu_3 \mu_4 \mu_1} \\ -T_{\mu_1 \mu_4 \mu_3 \mu_2} - T_{\mu_4 \mu_1 \mu_3 \mu_2} - T_{\mu_1 \mu_4 \mu_2 \mu_3} - T_{\mu_4 \mu_1 \mu_2 \mu_3} \\ +T_{\mu_3 \mu_4 \mu_1 \mu_2} + T_{\mu_4 \mu_3 \mu_1 \mu_2} + T_{\mu_3 \mu_4 \mu_2 \mu_1} + T_{\mu_4 \mu_3 \mu_2 \mu_1} \end{bmatrix}. \tag{28.21}
\end{aligned}$$

The factor $\frac{1}{12}$ is needed to obtain a projection operator ($Y^2 = Y$). (The easiest way to check this factor is to note that there are 16 terms in Y , but one finds the unit element only 12 times in Y^2 .)

The Young operators form a complete set of orthogonal projection operators. Denoting them by Y^J for (in this example) $J = 1, \dots, 10$, they satisfy

$$Y^J Y^K = 0 \text{ if } J \neq K; \quad Y^J Y^J = Y^J; \quad \sum_J Y^J = I. \tag{28.22}$$

They also give the irreps of the finite group $S(n)$ (the “symmetric group” of all $n!$ permutations of n objects), but that subject we do not discuss [2].

Let us now check whether the tensors obtained in (28.6) and (28.7) correspond to the states of an octet of $SU(3)$. We denote the states $|_1\rangle$, $|_2\rangle$ and $|_3\rangle$ of the **3** of $SU(3)$ by u, d and s , the usual notation for the three lightest quarks. Then the highest weight state of the decuplet **10** of $SU(3)$ is uuu , and by using the lowering operators $E_{-\alpha^1} = E_{12}^-$, $E_{-\alpha^2} = E_{23}^-$ and $E_{-\alpha^{13}}$ we get the rest of the decuplet. The lowering operators act as follows on the **3**,

$$\begin{array}{ccc}
d & \xleftarrow{E_{-\alpha^1}} & u \\
& \searrow E_{-\alpha^2} & \swarrow E_{-\alpha^{13}} \\
& s &
\end{array} \tag{28.23}$$

and using this, beginning with the state uuu one finds for the decuplet

$$\begin{array}{c}
 \begin{array}{ccc}
 & \xleftarrow{\frac{1}{\sqrt{3}}(udd+dud+ddu)} & \\
 ddd & \xleftarrow{\frac{1}{\sqrt{3}}(uud+udu+duu)} & uuu \\
 & \xleftarrow{\frac{1}{\sqrt{6}}(uds+usd+dus+dsu+sud+sdu)} & \\
 \frac{1}{\sqrt{3}}(dds+dsd+sdd) & \xleftarrow{\frac{1}{\sqrt{3}}(uus+usu+suu)} & \\
 & \xleftarrow{\frac{1}{\sqrt{3}}(uss+sus+ssu)} & \\
 & & sss
 \end{array}
 \end{array}
 \quad (28.24)$$

Moving to the left from uuu along the α^1 chain, by acting once with $E_{-\alpha^1}$, we find three states, uud , udu , duu , the totally symmetric combination of which, $\frac{1}{\sqrt{3}}(uud + udu + duu)$, belongs to the decuplet, leaving a two-dimensional space orthogonal to it which should contain the highest-weight states of the two octets. There is a two-parameter freedom to choose these highest weight states of the octets, but a useful way is to start with states which are antisymmetric in 2 quarks⁴, for example $\frac{1}{\sqrt{2}}(ud - du)u$ and $\frac{1}{\sqrt{2}}u(ud - du)$. (The third combination, antisymmetric in the first and last quark, $\frac{1}{\sqrt{2}}(uud - duu)$, is not linearly independent but equal to the sum of the previous two states.) By acting with the lowering operators on the highest weight state of the first octet, we find the complete first octet as follows

$$\begin{array}{c}
 \begin{array}{ccccc}
 & \frac{1}{\sqrt{2}}(ud - du)d & & \frac{1}{\sqrt{2}}(ud - du)u & \\
 & \swarrow & & \searrow & \\
 \frac{1}{\sqrt{2}}(ds - sd)d & & \frac{1}{2}[(us - su)d + (ds - sd)u] & & \frac{1}{\sqrt{2}}(us - su)u \\
 & \searrow & \bullet & \swarrow & \\
 & \frac{1}{\sqrt{2}}(sd - ds)s & & \frac{1}{\sqrt{2}}(us - su)s &
 \end{array}
 \end{array}
 \quad (28.25)$$

$I = 0$ state

The $I = 0$ state at the center should be annihilated by $E_{-\alpha^1}$ (or equivalently, be orthogonal to the $I = 1$ state in the center), and should also be orthogonal to the state in the center of the decuplet;

⁴The state $\frac{1}{\sqrt{2}}(ud - du)u$ is a linear combination of the states $T_{\mu_1\mu_2\mu_3}^{\square\square I}$ in (28.6) and $T_{\mu_1\mu_2\mu_3}^{\square\square II}$ in (28.7), namely $\frac{1}{3}(T_{112}^{\square\square I} + 2T_{112}^{\square\square II})$. Similarly, $u(ud - du)$ is equal to $\frac{1}{3}(T_{112}^{\square\square I} - T_{112}^{\square\square II})$.

this fixes it only as far as

$$\alpha(ud - du)s + \beta(us - su)d - \beta(ds - sd)u \quad (28.26)$$

However, using $E_{-\alpha^2}$ or $E_{-\alpha^{13}}$ one should find a linear combination of the two states in the center. For example, $E_{-\alpha^{13}}$ acting on $\frac{1}{\sqrt{2}}(ud - du)u$ produces a state proportional to $\frac{1}{\sqrt{2}}(sd - ds)u + \frac{1}{\sqrt{2}}(ud - du)s$, and this is only a linear combination of the state with $I = 1$ in the center and the state with $I = 0$ given in (28.26), if $\alpha = 2\beta$.

$$I = 0 \text{ state} = \frac{1}{\sqrt{12}}[2(ud - du)s + (us - su)d - (ds - sd)u] \quad (28.27)$$

Then $E_{-\alpha^2}$ acting on $\frac{1}{\sqrt{2}}(ud - du)d$ produces a linear combination of the state with $I = 0$ and $I = 1$ in the center. This fixes all states in the first octet.

We defined the states of the octet by acting with $E_{-\alpha^1}$ and $E_{-\alpha^2}$, starting with the highest weight state⁵. This yields a representation of $E_{-\alpha^1}$ and $E_{-\alpha^2}$ in terms of 8×8 matrices. However, one can also act with $E_{-\alpha^{13}}$ on states, for example acting on the state $\frac{1}{\sqrt{2}}(ud - du)d$ one obtains minus the state $\frac{1}{\sqrt{2}}(ds - sd)d$. This is not an inconsistency; it only means that in the 8×8 matrix representation of $E_{-\alpha^{13}}$ one finds an entry with -1 .

There is also the second octet which starts with the highest weight state $\frac{1}{\sqrt{2}}u(ud - du)$, and using the 3 lowering operators we can find the complete second octet as well. In all there are 27 states which fall into 4 multiplets of dimension **10**, **8**, **8**, **1**. This agrees with the number of 4 tensor multiplets in (28.8) with the same number of components as the number of corresponding states. We come now to the point we wanted to reach: to explicitly check that the tensor multiplets agree with the state multiplets. The tensors $T^{\square\square I}$ and $T^{\square\square II}$ have each 27 components, but most components vanish or are equal to other components, such that there are only 8 independent components for each. For example, starting with (28.7), and using the notation $\mu = 1 = u$, $\mu = 2 = d$ and $\mu = 3 = u$, we find from (28.7) an expression for T_{udu} , and similar expressions for T_{duu} and T_{uud}

$$T_{udu}^{\square\square I} = udu + udu - duu - duu \sim (ud - du)u; \quad T_{duu}^{\square\square I} = T_{udu}^{\square\square I}; \quad T_{uud}^{\square\square I} = 0. \quad (28.28)$$

So T_{udu} corresponds to the highest weight state of the first octet. Let us now see whether taking other values of μ^1, μ^2, μ^3 we do find the other states of the octet. For example, first T_{udd}

$$T_{udd}^{\square\square I} = udd + dud - 2ddu; \quad T_{dud}^{\square\square I} = 0; \quad T_{ddu}^{\square\square I} = -T_{udd}^{\square\square I}. \quad (28.29)$$

⁵Except the state at the lower right in (28.25) which was obtained by acting with $E_{-\alpha^{13}}$, but we obtain this same state if we act with $E_{-\alpha^2}$ on the two states in the center.

This is **not** a state of the multiplet. The same problem arises for $T^{\square II}$. Now one finds

$$\begin{aligned} T_{uud}^{\square II} &= 0; & T_{udu}^{\square II} &= 2udu - duu - uud; & T_{duu}^{\square II} &= -T_{udu}^{\square II} \\ T_{ddu}^{\square II} &= 0; & T_{dud}^{\square II} &= 2dud - udd - ddu; & T_{udd}^{\square II} &= -T_{dud}^{\square II} \end{aligned} \quad (28.30)$$

What is the solution of this conundrum?

The resolution is that one has constructed the wrong tensor multiplets. Rather than using the Young operator QP one should use⁶ PQ . So one should first antisymmetrize and then symmetrize the tensors. We leave as an exercise to start all over with these different multiplets, and check that they correspond precisely to the state multiplets.

Bibliography

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⁶I thank Howard Georgi (private communication) who ran into the same problem and found the solution given in the text.